EXHIBIT NYSLOF-1

Date of Request: June 14, 2019 Request No NYSLOF-4
Due Date: June 24, 2019 NG Request No. NG-659

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
Case Nos. 19-G-0309 & 19-G-0310
Gas Utilities Rates

Request for Information

FROM: New York State Laborers' Organizing Fund, Harrison Watkins

TO: National Grid, Gas Infrastructure and Operations Panel

SUBJECT: Gas Operations / Maintenance

Request:

Notes: All references to workpapers should be understood to include paper and electronic workpapers, computer-model input or output reports, internal memoranda, studies, or publications. In responding to requests for workpapers, the Company should provide both (1) workpapers created in the process of undertaking calculations; and (2) workpapers relied on to support input or analytical assumptions. When providing electronic spreadsheet files, please leave all cell formulas intact; i.e., do not range value cell formulas of spreadsheets provided. In addition, responses to requests for electronic spreadsheet files should include any linked electronic spreadsheets, as well as all other spreadsheet files or workpapers relied on to develop inputs for the requested spreadsheet file.

For the purposes of these information requests, "Significant Capital Programs" include the Proactive Main Replacement (LPP) Program, the Gas Safety Integrity Management and Integrity Verification: IMP and IVP Programs, the Transmission System Integrity Program, the city/state construction programs (CSC), the Metropolitan Reliability Infrastructure (MRI) Project, and the Renewable Natural Gas (RNG) Interconnections Program.

4. Please provide any workpapers or documents containing any analysis to forecast contractor crew levels for construction activities on Significant Capital Program projects.

Response:

KEDNY tracks historical crew counts and work completed to quantify crew productivity at the start of every year. Historic productivity for main and service installation is applied to the forecast work plan units to estimate crew count needs. See Attachment 1.

KEDLI forecasts contractor crew levels based on historical work completed and actual crew counts for prior years. This history is used to calculate average crew productivity by main and service activities and to estimate the average number of crews needed to complete the forecast work plan units in future fiscal years. See Attachment 2.

Name of Respondent:
Victor Vientos
Melissa Mancini

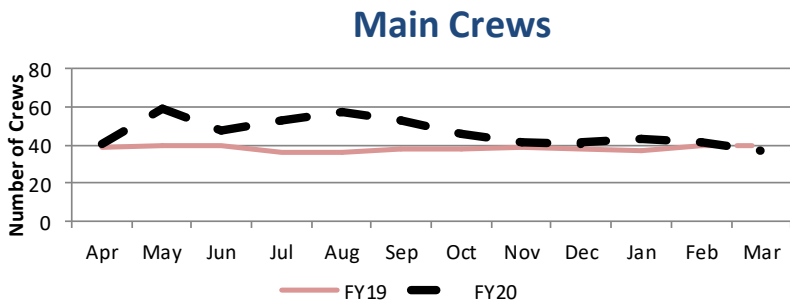
Date of Reply:
June 24, 2019

Resource Plan – [redacted] Main Crews

KEDNY

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Main Crews	Max.	Min.	Avg.
FY19 Actual Crew Counts	**40**	**36**	**38**
Encroachments	**0**	**0**	**0**
Gas Planning/RCV Programs	**1**	**0**	**0**
Gas System Reinforcement	**7**	**0**	**2**
Public Works	**34**	**20**	**26**
Install Main	**5**	**2**	**3**
Leak Prone Pipe	**17**	**10**	**13**
Pit Work / Test Holes	**3**	**3**	**3**
FY20 Anticipated Crew Needs	**60**	**37**	**47**



Methodology

- **All work plans built on a 5-day week**

Resources

- **Current workforce: 40 crews**
- **During peak time, utilize overtime / extended days to address resource gaps**
- **Daily Productivity Assumptions:**
  - **Blanket Assumption: 50ft per crew**

Risk to Plan

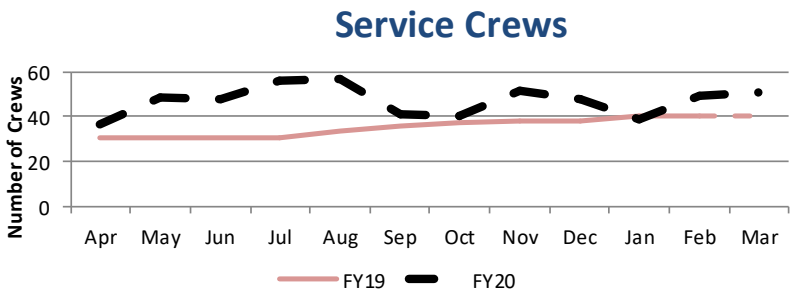
- **Working Stipulations on Permits**

Resource Plan – [redacted] Service Crews

KEDNY

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Service Crews	Max.	Min.	Avg.
FY19 Actual Crew Counts	**40**	**31**	**36**
Encroachments	**0**	**0**	**0**
Gas Planning/RCV Programs	**1**	**0**	**0**
Gas System Reinforcement	**3**	**0**	**1**
Local Law 30	**11**	**6**	**8**
Public Works	**24**	**13**	**19**
Leak Prone Pipe	**18**	**4**	**10**
Install Services	**10**	**6**	**8**
Maintain Assist	**1**	**0**	**1**
FY20 Anticipated Crew Needs	**56**	**36**	**46**



Methodology

- **All other programs built on a 5-day week**

Resources

- **Current Workforce: 40 crews**
- **During peak time, utilize overtime / extended days to address resource gaps**
- **Daily Productivity Assumptions:**
  - **Growth Services: 0.8 services/crew**
  - **D&R : 2.8 services/crew**
  - **Everything Else: 3.0 services/crew**

Risk to Plan

- **Working Stipulations on Permits**

All Programs - Planned Crew levels

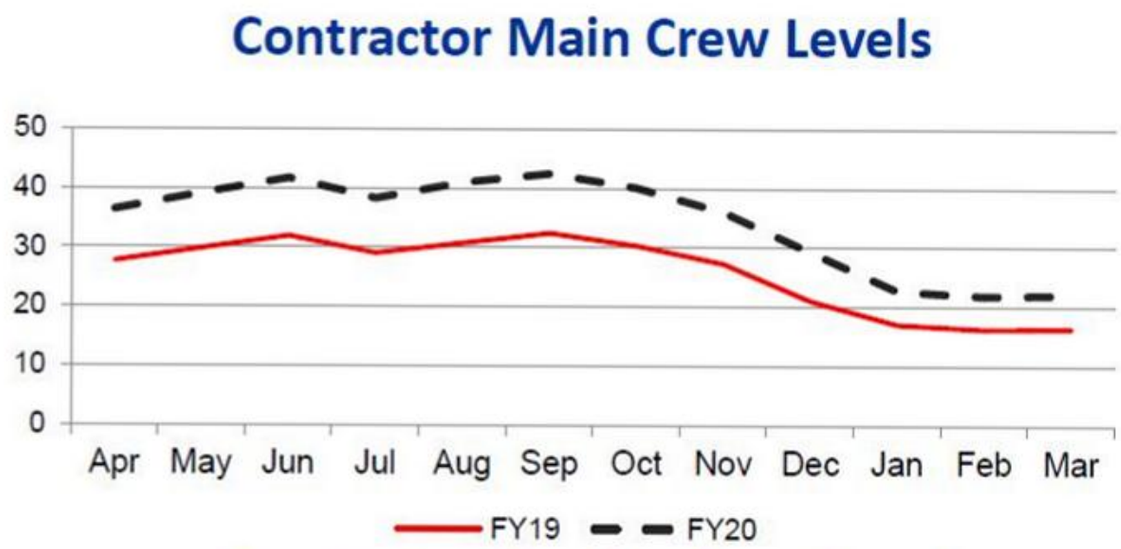
KEDLI

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Main	Max	Min	Average
FY 19 Totals	37	16	34
Base Growth	6	3	5
Reinforcement	6	0	3
Public Works	2	0	1
MRP	32	17	26
FY 20 Totals	42	22	34

1) Need 5 main crews at historical OT levels

2) Overtime & extended days reduces crew requirements



Service	Max	Min	Average
FY 19 Totals	53	29	44
Base Growth	12	7	9
Reinforcement	5	0	2
Public Works	1	0	0
MRP	47	18	37
FY 20 Totals	61	28	48

1) Need 8 service crews at historical OT levels

2) Contractors often can flex crews between main & service crews reducing risk

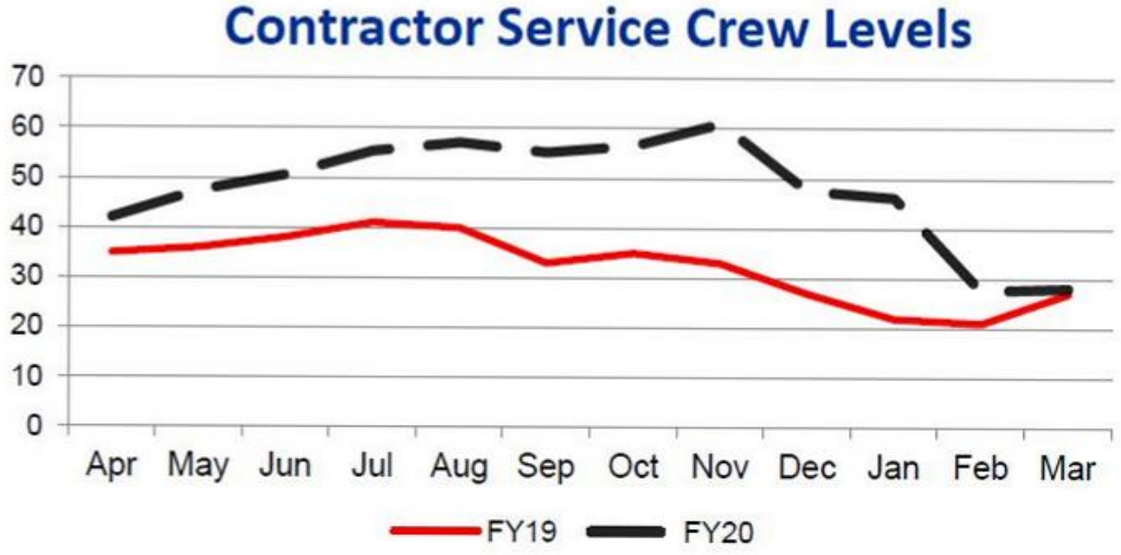


EXHIBIT NYSLOF-2

Date of Request: June 14, 2019 Request No NYSLOF-19
Due Date: June 24, 2019 NG Request No. NG-674

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
Case Nos. 19-G-0309 & 19-G-0310
Gas Utilities Rates

Request for Information

FROM: New York State Laborers' Organizing Fund, Harrison Watkins

TO: National Grid, Gas Infrastructure and Operations Panel

SUBJECT: Gas Operations/Maintenance

Request:

Notes: All references to workpapers should be understood to include paper and electronic workpapers, computer-model input or output reports, internal memoranda, studies, or publications. In responding to requests for workpapers, the Company should provide both (1) workpapers created in the process of undertaking calculations; and (2) workpapers relied on to support input or analytical assumptions. When providing electronic spreadsheet files, please leave all cell formulas intact; i.e., do not range value cell formulas of spreadsheets provided. In addition, responses to requests for electronic spreadsheet files should include any linked electronic spreadsheets, as well as all other spreadsheet files or workpapers relied on to develop inputs for the requested spreadsheet file.

For the purposes of these information requests, "Significant Capital Programs" include the Proactive Main Replacement (LPP) Program, the Gas Safety Integrity Management and Integrity Verification: IMP and IVP Programs, the Transmission System Integrity Program, the city/state construction programs (CSC), the Metropolitan Reliability Infrastructure (MRI) Project, and the Renewable Natural Gas (RNG) Interconnections Program.

19. Regarding the Company's Proactive Main Replacement Program (LPP):

a. Please provide the total LPP expenditures by year for calendar year (CY) 2016, CY2017, and CY2018.

i. For each year, please detail total expenditures paid to outside construction contractors by individual contractor.

ii. For each year, please detail the number of construction contractor FTEs employed on LPP projects by individual contractor.

b. For the rate year and data years, please detail what percentage of LPP expenditures will be for construction services procured from outside contractors.

c. Refer to GIOP-2, please detail the forecast LPP replacement unit costs by component for the rate year and data years.

Response:

a. See the Companies' response to DPS-084.

i. The tables below provides the total expenditures paid to outside construction contractors in calendar year (CY) 2016, CY2017, and CY2018 for KEDNY and KEDLI, respectively. The Company objects to the request for a breakout by individual contractor, as such request is not relevant to the issues in these proceedings.

Company	**CY16**	**CY17**	**CY18**
KEDNY	$35,644,816	$86,493,402	$123,160,629
KEDLI	$58,081,699	$68,104,720	$133,424,134

ii. The Companies do not track resources performing construction activities for capital programs in terms of FTEs.

b. See Attachments 1 and 2 which provides forecast contractor costs as a component of the Rate Year and Data Years LPP replacement unit costs.

c. See Attachments 1 and 2 which provide the LPP replacement unit costs by component in the Rate Year and Data Years for KEDNY and KEDLI, respectively.

Name of Respondent:
Mitchell Hoffman
Patricia McVeigh

Date of Reply:
June 24, 2019

			Unit Costs			
Company	**Program**	**L02 Cost Type**	**Rate Year**	**Data Year 1**	**Data Year 2**	**Data Year 3**
KEDNY	LPP	Base Labor	$ 48	$ 49	$ 51	$ 52
		Overtime	$ 32	$ 33	$ 34	$ 35
		Other Employee Benefit	$ 28	$ 29	$ 30	$ 30
		Pension and OPEB	$ 23	$ 24	$ 24	$ 25
		Employee Expenses	$ 0	$ 0	$ 0	$ 0
		Capital Overheads	$ 132	$ 136	$ 139	$ 142
		Consultants	$ 1	$ 1	$ 1	$ 1
		Contractors	$ 949	$ 973	$ 997	$ 1,021
		Capitalized Interest	$ 8	$ 8	$ 8	$ 9
		Materials	$ 52	$ 53	$ 55	$ 56
		Operating Revenue	$ (0)	$ (0)	$ (0)	$ (0)
		Transportation	$ 6	$ 6	$ 6	$ 7
			$ 1,280	**$ 1,312**	**$ 1,345**	**$ 1,378**

			Unit Costs			
Company	**Program**	**L02 Cost Type**	**Rate Year**	**Data Year 1**	**Data Year 2**	**Data Year 3**
KEDLI	LPP	Base Labor	$ 12	$ 13	$ 13	$ 13
		Overtime	$ 7	$ 7	$ 7	$ 7
		Other Employee Benefit	$ 6	$ 6	$ 6	$ 7
		Pension and OPEB	$ 4	$ 5	$ 5	$ 5
		Employee Expenses	$ 0	$ 0	$ 0	$ 0
		Capital Overheads	$ 71	$ 73	$ 75	$ 77
		Consultants	$ 4	$ 4	$ 4	$ 4
		Contractors	$ 195	$ 200	$ 205	$ 210
		Capitalized Interest	$ 4	$ 4	$ 4	$ 5
		Materials	$ 6	$ 6	$ 7	$ 7
		Operating Revenue	$ -	$ -	$ -	$ -
		Transportation	$ 5	$ 5	$ 5	$ 5
			$ 298	**$ 305**	**$ 313**	**$ 321**

EXHIBIT NYSLOF-3

Date of Request: June 14, 2019 Request No NYSLOF-8
Due Date: June 24, 2019 NG Request No. NG-663

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
Case Nos. 19-G-0309 & 19-G-0310
Gas Utilities Rates

Request for Information

FROM: New York State Laborers' Organizing Fund, Harrison Watkins

TO: National Grid, Gas Infrastructure and Operations Panel

SUBJECT: Gas Operations / Maintenance

Request:

Notes: All references to workpapers should be understood to include paper and electronic workpapers, computer-model input or output reports, internal memoranda, studies, or publications. In responding to requests for workpapers, the Company should provide both (1) workpapers created in the process of undertaking calculations; and (2) workpapers relied on to support input or analytical assumptions. When providing electronic spreadsheet files, please leave all cell formulas intact; i.e., do not range value cell formulas of spreadsheets provided. In addition, responses to requests for electronic spreadsheet files should include any linked electronic spreadsheets, as well as all other spreadsheet files or workpapers relied on to develop inputs for the requested spreadsheet file.

For the purposes of these information requests, "Significant Capital Programs" include the Proactive Main Replacement (LPP) Program, the Gas Safety Integrity Management and Integrity Verification: IMP and IVP Programs, the Transmission System Integrity Program, the city/state construction programs (CSC), the Metropolitan Reliability Infrastructure (MRI) Project, and the Renewable Natural Gas (RNG) Interconnections Program.

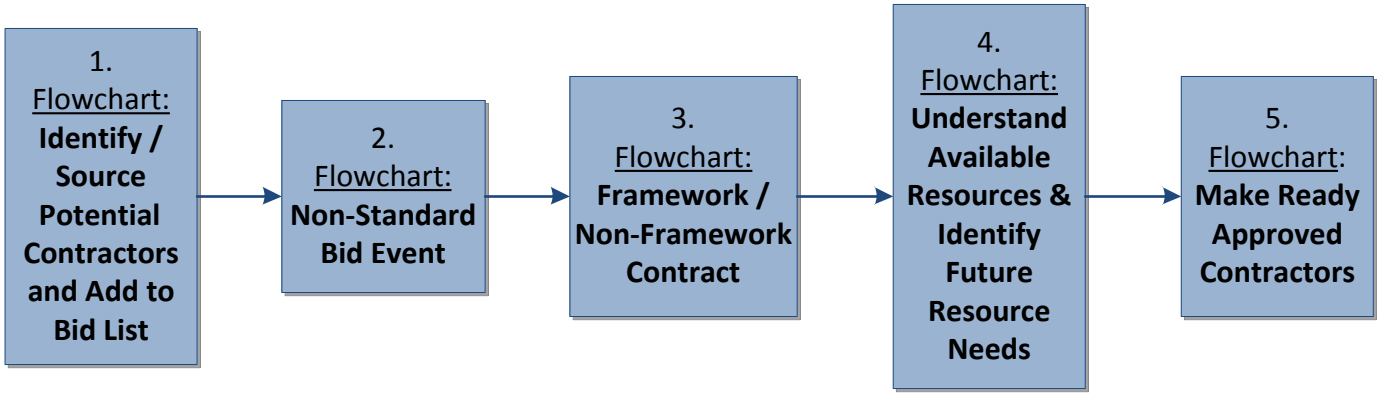
8. Please detail the process by which construction contractors are qualified and invited to bid on construction projects for the Company's Significant Capital Program projects. Please provide a list of contractors who have met these qualification standards.

Response:

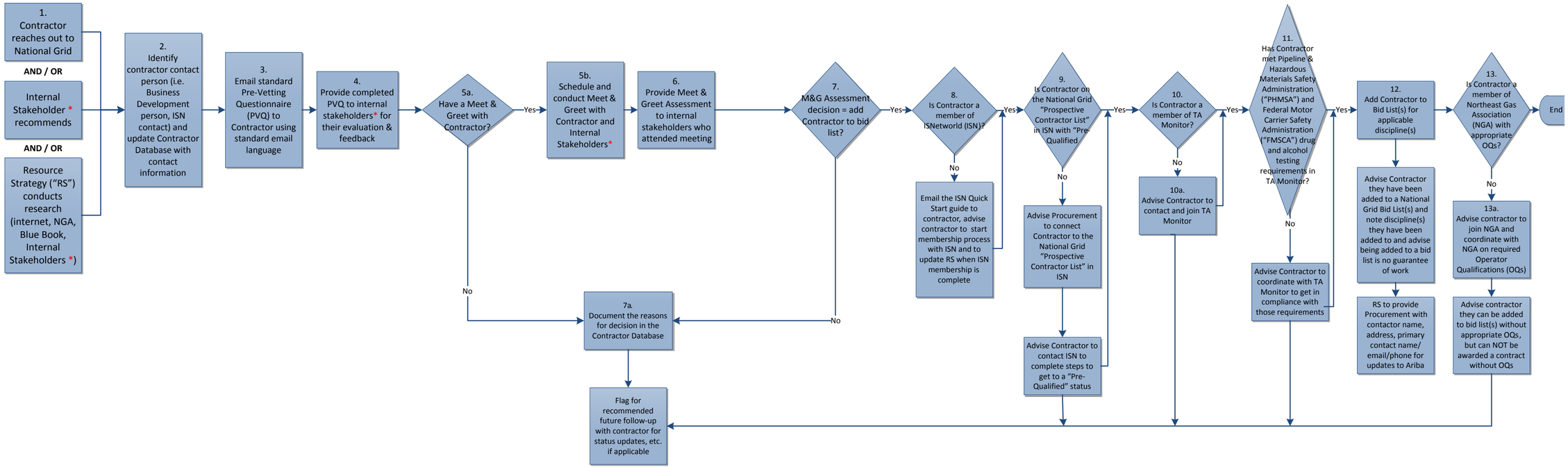
Attachment 1 is the Sourcing and Qualifying Contractors Process Flowchart. The flowchart summarizes the steps taken to qualify contractors for work. The Company objects to the request for a list of contractors who have met these qualifications. Such a list is proprietary and not relevant to these proceedings.

Name of Respondent:	Date of Reply:
James Lennon	June 24, 2019

Resource Strategy: Overview of Process Flowcharts (Ctrl+Click on boxes below for links to flowchart)



Identify / Source Potential Contractors and Add to Bid List



* Internal Stakeholders: Procurement, Construction, Project Management, Project Development, Operations, etc.

EXHIBIT NYSLOF-4

Date of Request: June 14, 2019 Request No NYSLOF-9
Due Date: June 24, 2019 NG Request No. NG-664

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
Case Nos. 19-G-0309 & 19-G-0310
Gas Utilities Rates

Request for Information

FROM: New York State Laborers' Organizing Fund, Harrison Watkins

TO: National Grid, Gas Infrastructure and Operations Panel

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9. Please provide copies of any documents used by the Company to assess the qualification of external construction contractors or to compare construction contractors as it relates to the contractor's capacity to meet all safety, regulatory, engineering, environmental, legal, manpower, and financial requirements.

Response:

Please see the following attachments:

- Attachment 1 - National Grid Terms and Conditions for gas bid work

- Attachment 2 - Scoring Template used to evaluate the technical and commercial proposals of the bidders

- Attachment 3 - The Northeast Gas Associations list of Operator Qualifications used to assess individual gas laborers' ability to perform specific disciplines of work

- Attachment 4 - Gas Services Vetting Questionnaire used to assess a contractor's basic ability to perform a variety of gas work

Name of Respondent:
James Lennon

Date of Reply:
June 24, 2019

nationalgrid

Gas Services – Prospective Supplier Questionnaire

1. The information contained within this Prospective Supplier Questionnaire is confidential and proprietary to National Grid, and is to be used by the recipient solely for the purpose of responding to this Questionnaire.

   This Questionnaire does not constitute an offer by National Grid to enter into a contract, nor does any response constitute an acceptance of an offer, nor does it bind National Grid in any way.

2. It is understood that the specific work would involve the following:

   Perform gas services: This work will include the use of contractor's equipment (or other agreed upon methods) to perform gas services work per industry standards at specified locations within National Grid's franchise territories satisfying requirements of applicable laws consistent with damage prevention practices and procedures.

3. Responses to this Questionnaire will be evaluated by National Grid's Contractor Resource Team with a focus on the following key factors:
   a. Experience
   b. Approach
   c. Qualifications of staff
   d. Resource availability
   e. Equipment
   f. Safety culture

4. **Basic Company Information:** Please provide the following:
   - Date completed:
   - Company name:
   - Primary contact (Name & Title):
   - Phone #:
   - Email:
   - Website address:
   - Address of primary company location address (Street, City, State, Zip):
   - # of employees:
   - Years in business:
   - Are you a Union company?
   - Union Affiliation(s):
   - Subsidiary or parent companies (if applicable):

5. **Safety:** Please discuss your firm's safety record and environmental initiatives.

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Gas Services – Prospective Supplier Questionnaire

6. **Personnel/Operator Qualifications:** Please discuss the Personnel/Operator Qualification Program(s) your firm currently has in place:

   Are you a member of Northeast Gas Association (NGA)?

   Do you use NGA for your Operator Qualifications (OQs)?

   Are you a member of any other gas association?

   If YES, please specify other gas association:

7. **Capabilities:** Do you currently have the capability to perform gas services in these areas? Indicate YES or NO for each region below:

   MA:
   RI*:
   Upstate NY:
   New York City*:
   Long Island NY*:

   * Must be UNION to work in these regions

   Please list other states/countries you currently perform these services in:

   How many gas projects do you currently have in progress and in which locations?

   Please indicate the name, and location, of other gas utility companies your firm has provided services to:

   How many gas crews (by crew type) are expected to be deployed across your footprint over the next 12 months? (Please indicate if these crews are NGA qualified gas crews).

8. **Please provide a specific description of the type of gas services the company is interested in providing to National Grid:**

9. For the gas services your company is interested in providing to National Grid, please describe your firm's past and current experience with gas projects similar to the work described below.

   Please specify below whether the company has the capability to SELF PERFORM the work directly (or whether you SUB CONTRACT). Indicate "N/A" for any items below if not applicable to the services you are interested in providing to National Grid.

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Gas Services – Prospective Supplier Questionnaire

Gas Construction: Mains & Services: _____

Gas Construction: Growth Services: _____

Gas Construction: Local Law 30 (New York City only): _____

Gas Construction: Article VII (New York only): _____

Gas Construction: Pressure Regulator Station (Rehab, Replacement, New Construction): _____

_____

Gas Construction: Transmission Pipeline: _____

Gas Construction: Distribution Pipeline: _____

Gas Construction: Residential Underground Development: _____

Excavation: Deep Pit Excavation: _____

Excavation: Misc. Excavation: _____

Traffic Control & Flagging: _____

Traffic Loop Repair: _____

Paving & Restoration: _____

Landscape Restoration: _____

Saw Cutting & Coring: _____

Vacuum Excavation: _____

Gas Pipe Lining: _____

Gas Pipe Sealing: _____

Locate & Mark Out: _____

LNG / CNG Construction: _____

nationalgrid

Gas Services – Prospective Supplier Questionnaire

Gas Engineering Services: _____

Gas Leak Survey: _____

Gas Pinpointing: _____

Gas Leak Repair with Keyhole Technology: _____

Gas Corrosion Detection / Mitigation: _____

Cross Bore Services: _____

Inside Gas Service Inspections: _____

Non-Destructive Testing / Evaluation: _____

Certified Welding Inspections: _____

Horizontal Directional Drilling: _____

Gas Line Welding: _____

Crane & Rigging: _____

Gas Pigging Inspections: _____

Other related Gas Services: _____

10. **ISNetworld**: Are you a member of ISNetworld (www.isnetworld.com)? If yes, please provide your member#:

    If not a member of ISN, are you willing to join ISN?
    (*Please note that in order to be added to a National Grid bid list, you need to be a member of ISNetworld).*

    What is your current EMR rating?

11. **Equipment:** Please list the equipment your firm deploys related to the provision of gas services. If needed, attach additional documentation.

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Gas Services – Prospective Supplier Questionnaire

12. **Diversity:** Is your company certified as a Minority or Equal Partner Program Supplier?

If yes, please indicate which category(s) you are certified in and attach your certification documents.

Does your company sub-contract work to certified companies?

13. **Bonding/Insurability:** Please provide your company's current bonding/insurability amount:

EXHIBIT NYSLOF-5

Date of Request: June 14, 2019 Request No NYSLOF-9
Due Date: June 24, 2019 NG Request No. NG-664

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
Case Nos. 19-G-0309 & 19-G-0310
Gas Utilities Rates

Request for Information

FROM: New York State Laborers' Organizing Fund, Harrison Watkins

TO: National Grid, Gas Infrastructure and Operations Panel

SUBJECT: Gas Operations / Maintenance

Request:

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9. Please provide copies of any documents used by the Company to assess the qualification of external construction contractors or to compare construction contractors as it relates to the contractor's capacity to meet all safety, regulatory, engineering, environmental, legal, manpower, and financial requirements.

Response:

Please see the following attachments:

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- Attachment 2 - Scoring Template used to evaluate the technical and commercial proposals of the bidders

- Attachment 3 - The Northeast Gas Associations list of Operator Qualifications used to assess individual gas laborers' ability to perform specific disciplines of work

- Attachment 4 - Gas Services Vetting Questionnaire used to assess a contractor's basic ability to perform a variety of gas work

Name of Respondent:
James Lennon

Date of Reply:
June 24, 2019

"Project Name" Bid Evaluation Form

RATING

0 = No Response
1 = Does not Meet Requirements
2 = Meets Some Requirements
3 = Meets All Requirements
4 = Exceeds Some Requirements
5 = Exceeds All Requirements

CRITERIA	Weighting	Supplier Name #1			Supplier Name #2		
Technical Capability	Weight, %	Rating	Weighted Score	Comments	Rating	Weighted Score	Comments
Viability							
Proposed construction method	20%		0.00			0.00	
Schedule							
Proposed schedule meets or exceeds requirement	50%		0.00			0.00	
Pipeline Operator Qualifications (OpQual)							
Has the NGA OpQual program in place	20%		0.00			0.00	
Adherance to Technical Specifications							
Acceptance with no exceptions	10%		0.00			0.00	
TOTAL Technical Capability	100%		0.00			0.00	
Resource Capability	Weight, %	Rating	Weighted Score	Comments	Rating	Weighted Score	Comments
Existing employees							
Number of available crews	50%		0.00			0.00	
Bench strength							
Additional crews able to call in for expanded work	20%		0.00			0.00	
Existing workload							
Ability to do multiple projects based on existing workload	20%		0.00			0.00	
Subcontractors							
Subcontractors provided are acceptable	10%		0.00			0.00	
TOTAL Resource Capability	100%		0.00			0.00	
Experience	Weight, %	Rating	Weighted Score	Comments	Rating	Weighted Score	Comments
Pipeline Construction							
General experience	50%		0.00			0.00	
National Grid							
Recent experience working with National Grid	25%		0.00			0.00	
Safety							
Safety Record	15%		0.00			0.00	
Environmental							
Envioronmental Record	10%		0.00			0.00	
TOTAL Experience	100%		0.00			0.00	
Commercial_ Price	Weight, %	Rating	Weighted Score	Comments	Rating	Weighted Score	Comments
Lump Sum Price							
Competitive project pricing (lower $ = higher score)	75%		0.00			0.00	
Unit/ T&E Rates							
Lower $ = higher score	15%		0.00			0.00	
Exceptions							
Exceptions to commercial terms (e.g., T's &C's)	10%		0.00			0.00	
TOTAL Commercial_ Price	100%		0.00			0.00	

EXHIBIT NYSLOF-6

Date of Request: June 14, 2019 Request No NYSLOF-10
Due Date: June 24, 2019 NG Request No. NG-665

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
Case Nos. 19-G-0309 & 19-G-0310
Gas Utilities Rates

Request for Information

FROM: New York State Laborers' Organizing Fund, Harrison Watkins

TO: National Grid, Gas Infrastructure and Operations Panel

SUBJECT: Gas Operations / Maintenance

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10. Please list all workforce safety, worker compensation (eg. EMR scores), health and safety, safety and reliability, environmental, productivity, or performance metrics used by the Company to evaluate or qualify outside construction contractors. Please include the methodology to calculate each individual metric.

Response:

Please see the following attachments for metrics used by the Company's Procurement function to evaluate contractors:

- Attachment 1 - ISNetworld requirements

- Attachment 2 - National Grid Gas Construction Contractor Safety and Compliance Metrics
- Attachment 3 - N-1402 National Grid Contractor Safety Requirements
- Attachment 4 - National Grid New Outstanding Onboarding Questionnaire
- Attachment 5 - National Grid Showstopper Grading Questionnaire
- Attachment 6 - National Grid Written Safety Program Requirement Matrix
- Attachment 7 - National Grid Contractor Insurance Requirements

Name of Respondent:
James Lennon

Date of Reply:
June 24, 2019

Starting at a high level, contractors are asked to achieve an acceptable grade (A,B,C) for National Grid based on National Grid specific requirements.

Contractors who achieve an A (90-100), B (80-89.99) or C (70-79.99) grade are considered 'Acceptable', while anyone below that threshold (D or F grade) is considered unacceptable and requires additional approval.

We have detailed the two showstopper requirements (must be accepted to perform work) along with the remaining 4 items that combine to create the 100 point scale used to measure compliance for National Grid below.

1. Contractor Evaluation Criteria tracked in ISNetworld:

- **Insurance** *(Showstopper)* – If contractor has not submitted acceptable insurance documentation, they will automatically drop to an unapproved grade (F)
    - Requirements built with National Grid based on standard MSA obligations and are attached for your review
    - Workers Compensation – Required as a part of the insurance requirements and reviewed against National Grid and industry standards, outlined in the insurance requirement attachment
        - If any portion of a contractors insurance is not accepted based on the review criteria, the entire document will be rejected and the contractor given specific review notes on what needs to be updated
            - *Contractor receives -100 points if document is not submitted, expired or rejected and 0/0 points if accepted

- **N-1402 Contractor Safety Form Acknowledgement** *(Showstopper)* – If not acknowledged, contractor automatically dropped to an unapproved grade (F)
    - Contractors given 60+ day grace period to acknowledge before grades were impacted
        - *Contractor receives -100 points if document is not acknowledged and 0/0 points if it has been

- **Showstopper Grading** *(33% of contractor overall score)* – National Grid Graded Statistics comprised of 4 metrics:
    - TRIR (Calculation below)
    - DART (Calculation below)
    - Citations (OSHA, MSHA or EPA)
    - Fatalities (Previous 3 years, pulled from verified OSHA forms)
    - Metric Scoring
        - 90-100% yields 33/33 Points
        - 80-89.99% yields 22 Points
        - 60-79.99% yields 10 Points
        - 40-59.99% yields 5 Points
        - 11-39.99% yields 2 Points
        - Incomplete Data yields 0 points
    - Some questions are included for visibility only, while some are graded – listed in Column E (attachment #3)

- **New Vendor Onboarding Application Form** *(30% of contractor overall score)* – Questionnaire developed with National Grid asking contractors to provide responses to standard Health and Safety Questions.
    - Metric Scoring:
        - 90-100% acceptable answers yield 30/30 points

        - 80-89.99% yields 20 Points
        - 70-79.99% yields 10 Points
        - 0.1-69.99% yields 2 Points
    - Some questions are included for visibility only, while some are graded – listed in Column C (attachment #4)

- **Written Program Review** *(25% of contractor overall score)* – Written Health and Safety programs submitted by the contractor based on the work type(s) they have selected in their account
    - Requirements developed with National Grid for each work type (outlined in the attached matrix)
    - Contractor submitted programs are reviewed against OSHA standard requirements by the ISN review team
    - Metric Scoring:
        - 90-100% of programs required by National Grid accepted, yields 25 out of 25 points in this section.
        - 80-89.99% Accepted Programs yields 20 Points
        - 70-79.99% Accepted Programs yields 15 Points
        - 60-69.99% Accepted Programs yields 10 Points

- **Current Year Experience Modifier (EMR)** *(12% of contractor overall score)* – scored separately from insurance and workers compensation requirements
    - Submitted by contractor and reviewed by the ISN review team
    - Metric Scoring:
        - An EMR Less than 1 yields 12 Points
        - An EMR between 1.00 and 1.5 yields 9 Points
        - An EMR greater than 1.5 yields 2 points

2. Calculations:

All rates calculated are compared to the contractors BLS (Bureau of Labor Statistics) industry average to determine performance relative to peers and subsequent point allocation from National Grid.

- **Total Recordable Incident Rate [TRIR]** represents the number of recordable injuries and illnesses per 100 full-time workers and is calculated as:
    - **TRIR = (N/EH) x 200,000**
    - **Where:**
        - N = Total number of recordable injuries and illnesses.
        - EH = Total exposure hours, i.e. total hours worked by all employees during the calendar year.
        - 200,000 = Base for 100 equivalent full-time workers (working 40 hours per week, 50 weeks per year).

- **Days Away, Restricted Activity or Job Transfer Incident Rate [DART]** represents the number of recordable nonfatal injuries and illnesses resulting in days away from work, restricted activity or job transfer per 100 full-time workers and is calculated as:
    - **DART = [(N_{DAFW} + N_{DRAJT})/EH] x 200,000**
    - **Where:**
        - N_{DAFW} = Total number of nonfatal recordable injuries and illnesses resulting in days away from work.
        - N_{DRAJT} = Total number of nonfatal recordable injuries and illnesses resulting in restricted activity or job transfer.
        - EH = Total exposure hours, i.e. total hours worked by all employees during the calendar year.
        - 200,000 = Base for 100 equivalent full-time workers (working 40 hours per week, 50 weeks per year).

EXHIBIT NYSLOF-7

Date of Request: June 14, 2019 Request No NYSLOF-10
Due Date: June 24, 2019 NG Request No. NG-665

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
Case Nos. 19-G-0309 & 19-G-0310
Gas Utilities Rates

Request for Information

FROM: New York State Laborers' Organizing Fund, Harrison Watkins

TO: National Grid, Gas Infrastructure and Operations Panel

SUBJECT: Gas Operations / Maintenance

Request:

Notes: All references to workpapers should be understood to include paper and electronic workpapers, computer-model input or output reports, internal memoranda, studies, or publications. In responding to requests for workpapers, the Company should provide both (1) workpapers created in the process of undertaking calculations; and (2) workpapers relied on to support input or analytical assumptions. When providing electronic spreadsheet files, please leave all cell formulas intact; i.e., do not range value cell formulas of spreadsheets provided. In addition, responses to requests for electronic spreadsheet files should include any linked electronic spreadsheets, as well as all other spreadsheet files or workpapers relied on to develop inputs for the requested spreadsheet file.

For the purposes of these information requests, "Significant Capital Programs" include the Proactive Main Replacement (LPP) Program, the Gas Safety Integrity Management and Integrity Verification: IMP and IVP Programs, the Transmission System Integrity Program, the city/state construction programs (CSC), the Metropolitan Reliability Infrastructure (MRI) Project, and the Renewable Natural Gas (RNG) Interconnections Program.

10. Please list all workforce safety, worker compensation (eg. EMR scores), health and safety, safety and reliability, environmental, productivity, or performance metrics used by the Company to evaluate or qualify outside construction contractors. Please include the methodology to calculate each individual metric.

Response:

Please see the following attachments for metrics used by the Company's Procurement function to evaluate contractors:

- Attachment 1 - ISNetworld requirements

- Attachment 2 - National Grid Gas Construction Contractor Safety and Compliance Metrics
- Attachment 3 - N-1402 National Grid Contractor Safety Requirements
- Attachment 4 - National Grid New Outstanding Onboarding Questionnaire
- Attachment 5 - National Grid Showstopper Grading Questionnaire
- Attachment 6 - National Grid Written Safety Program Requirement Matrix
- Attachment 7 - National Grid Contractor Insurance Requirements

Name of Respondent:
James Lennon

Date of Reply:
June 24, 2019

Performance Metric	Topic	Methodology
Safety	Compliance Assessments	Total Safe or Compliant observations/Total Observations
	Contractor Incidents	# of Incidents reported/Miles of pipe installed; # of Incidents reported/1000's of One Call Tickets
	DART	DART = [(NDAFW + NDRAJT)/EH] x 200,000 Where: NDAFW = Total number of nonfatal recordable injuries and illnesses resulting in days away from work. NDRAJT = Total number of nonfatal recordable injuries and illnesses resulting in restricted activity or job transfer. EH = Total exposure hours, i.e. total hours worked by all employees during the calendar year. 200,000 = Base for 100 equivalent full-time workers (working 40 hours per week, 50 weeks per year).
Performance	Damage to Network (Self-Inflicted)	Self-Inflicted damages/Miles of pipe installed;
	Damage to Network (Result of Mismark)	Damages as a result of a Locator Mismark/1000's of One Call Tickets
Compliance	Quality Assurance/ Quality Control	Total Compliant observations/Total Observations
	Compliance Assessments	Total Safe or Compliant observations/Total Observations

EXHIBIT NYSLOF-8

Date of Request: June 14, 2019 Request No NYSLOF-10
Due Date: June 24, 2019 NG Request No. NG-665

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
Case Nos. 19-G-0309 & 19-G-0310
Gas Utilities Rates

Request for Information

FROM: New York State Laborers' Organizing Fund, Harrison Watkins

TO: National Grid, Gas Infrastructure and Operations Panel

SUBJECT: Gas Operations / Maintenance

Request:

Notes: All references to workpapers should be understood to include paper and electronic workpapers, computer-model input or output reports, internal memoranda, studies, or publications. In responding to requests for workpapers, the Company should provide both (1) workpapers created in the process of undertaking calculations; and (2) workpapers relied on to support input or analytical assumptions. When providing electronic spreadsheet files, please leave all cell formulas intact; i.e., do not range value cell formulas of spreadsheets provided. In addition, responses to requests for electronic spreadsheet files should include any linked electronic spreadsheets, as well as all other spreadsheet files or workpapers relied on to develop inputs for the requested spreadsheet file.

For the purposes of these information requests, "Significant Capital Programs" include the Proactive Main Replacement (LPP) Program, the Gas Safety Integrity Management and Integrity Verification: IMP and IVP Programs, the Transmission System Integrity Program, the city/state construction programs (CSC), the Metropolitan Reliability Infrastructure (MRI) Project, and the Renewable Natural Gas (RNG) Interconnections Program.

10. Please list all workforce safety, worker compensation (eg. EMR scores), health and safety, safety and reliability, environmental, productivity, or performance metrics used by the Company to evaluate or qualify outside construction contractors. Please include the methodology to calculate each individual metric.

Response:

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- Attachment 4 - National Grid New Outstanding Onboarding Questionnaire
- Attachment 5 - National Grid Showstopper Grading Questionnaire
- Attachment 6 - National Grid Written Safety Program Requirement Matrix
- Attachment 7 - National Grid Contractor Insurance Requirements

Name of Respondent:
James Lennon

Date of Reply:
June 24, 2019

FOREWORD

National Grid's vision is to be a world-class safety organization with zero injuries every day. This includes working to help ensure the safety of our employees, contractors and the community. National Grid is committed to delivering operational excellence, including excellent levels of safety internally and in cooperation with the external contractors we rely on.

The Executive Safety Committee provides review and input for Safety Policies and Procedures through the Safety Policies and Procedures Subcommittee.

The Safety department is the owner of this procedure and is responsible for maintaining and implementing this procedure, soliciting comments from stakeholders and revising as necessary.

This document, "Contractor Safety Requirements", represents the current contractor safety requirements that are unique to operations and various functional groups at National Grid. This document does not specifically reference actions that are required by OSHA, other laws, rules, or regulations. These are requirements that should be understood by the contractor and contractor compliance with all applicable federal, state and local laws, rules, and regulations is expected by National Grid as a contractual condition.

Questions regarding this procedure should be referred to the National Grid Safety Department.

This document will be updated as necessary to communicate all aspects of National Grid's contractor safety to bidders, current contractors and to reflect changes in National Grid's Safety Policies and Procedures.

Date of Review/Revision:

Revision	Date	Description
1	08/5/2004	Initial
2	3/2/2005	Additions
3	01/30/2007	Additions
4	08/01/2008	Additions
5	08/01/2010	Additions
6	02/01/2011	Audit recommendations included

Date of Review/Revision Continued:

Revision	Date	Description
7	09/11/2013	Additions included OH; technical changes; Compliance Monitoring; Ethics; Job Briefs
8	11/02/2015	Additions include Audit & IA recommendations; ISN alignment; technical changes, 1910. 269 updates
9	8/17/2016	Format update and technical changes
10	3/29/2017	Additions to sections 2.2.6 and 6.5
11	2/26/2018	Process Safety, PM&CC Electric and PM&CC Vegetation Additions
12	3/12/2019	Contaminated Site Work Additions

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1.0 CONTRACTOR SAFETY AT NATIONAL GRID

1.1 Definitions

Adverse Public Impact
Incident that disrupts service to the public or results in adverse public reaction.

Bulk Commodity Transportation
Activities involved in the movement of bulk commodities via truck, rail, plane or water vessel onsite and offsite on behalf of National Grid that if released could have safety and / or environmental consequences. Examples include but are not limited to: gasoline, oil, boiler chemicals, LNG, Nitrogen.

Compliance Assessments (CAs)
An act of observing and engaging in discussion with employees at a job site or work area locations. Compliance Assessments are documented using the Compliance Assessment checklist for each segment of operation and are not considered anonymous. Compliance Assessments are utilized to comply with internal policy and external regulatory requirements.

Contracted Services
Contracted Services refers to any activity that is conducted by an organization or individual under the terms of a purchase order. Contracted services may include all types of construction and maintenance services, tree trimming, building maintenance and demolition, electrical structure dismantling, site restoration, engineering design, recycling and waste disposal, drilling, rigging, electrical, and utility pole/structure maintenance.

Contractor
An independent person or company that undertakes a contract to provide materials or labor to perform a service or do a job and are responsible for the safety of his/her employees and subcontractors.

Contractor Orientation
Contractor orientation is intended to serve as a resource in order to provide the contractor with the tools necessary to educate their employees and subcontractors. The session is not intended to train the contractor management, their employees or subcontractors. The extent and content of the orientation session shall be commensurate with the scope and type of the contractor's activities.

Hazardous Conditions
A condition that can and is rectified immediately by the person who identified the hazard.

Incident
An unplanned event that has a human component and results in or could potentially result in harm to people, damage to property and/or adverse public impact.

Incident Management System (IMS)
National Grid's online incident management tool that allows the company to report safety, environmental and asset-related incidents, perform incident analysis, generate safety reports and monitor the organization's safety performance.

ISNetworld, Inc. (ISN or ISN System)
Third party contractor that is a global resource for connecting Hiring Clients with safe and reliable contractors and is a contractor information management system currently contracted with National Grid.

Job Brief
A planned interactive discussion with all contractor employees, sub-contractors and team members working on a job that occurs just prior to a job, task, project, prior to a change in planned work or when a new person arrives on site.

Major Hazard Asset (MHA)
A class of assets at National Grid, including Compressed Natural Gas (CNG), Gas Transmission (≥ 125 psig), Power Generation sites, Liquefied Natural Gas (LNG) plants, and LNG Trucking, in which any condition, or set of conditions, presents potential for a major accident to occur. Also referred to as process safety assets.

Major Accident
An event involving the release of potentially dangerous materials, the sudden and uncontrolled release of large amounts of energy (such as fires and explosions), or both. These can have catastrophic effects and can result in multiple injuries and fatalities, as well as substantial reputational, economic, property, and environmental damage

Operator Qualification (OQ)
As defined in the Code of Federal Regulations, Transportation, 49 Subpart 192.801 through 192.809 and/or DOT pipeline qualified for gas contractors doing work at National Grid. Additional state requirements pursuant to the state the contractor is working may be required.

Process Safety Management
Method of focusing and mitigating concerns of major hazards impacting safety, environmental damage and business losses. It is an organized effort to identify

and analyze the significance of hazardous situations associated with a process or activity to aid management in making critical safety decisions

Project Representative
National Grid Owner's Representative or designee who is assigned to certain contracted projects and communicates regularly with the contractor during the course of the contracted service. This person also ensures the work is being performed in accordance with the contract, including the safety requirements.

Purchase Order (P.O.)
An agreement/contract between National Grid and a contractor to provide services and/or materials. The P.O. is set up by Procurement. The term "Contract" and "P.O." are similar and may be used interchangeably. A "Blanket P.O." is set up for contractors whose work is on-going. A "one-time P.O." is set up for project work.

Qualified Electrical Worker
Those who are knowledgeable in the construction and operation of the electric power generation, transmission and/or distribution equipment involved, along with the associated hazards.

Qualified Gas Worker
Any contractor who performs covered tasks in accordance with National Grid's Operator Qualification Program and the Northeast Gas Association are required to be knowledgeable and meet all regulatory standards.

Risk Assessment
A risk assessment is the process of identifying hazards and calculating or ranking the associated risks according to: the likelihood of occurrence, the severity of the harm from the hazard, and the amount of time of exposure to the hazard.

Effective Safety Discussion (ESD)
A discussion with an individual or group about their safety programs, issues or concerns (safety plans, tools, equipment, procedures, etc.). They are safety discussions amongst employees that share similar work environments…office to office, field to field.

Safety Observer
A person who is responsible for alerting the work team to any potential unsafe conditions or lack of compliance with approved work practices, procedures or guidelines.

Transportation Advisor
Third party agency specializing in federal and company mandated drug and alcohol testing programs.

1.2 Introduction

Safety performance is a prime consideration in the selection of contractors. National Grid will stipulate safety performance requirements and responsibilities in our contracts, purchase orders (POs) and will hold the contractor accountable for meeting the contractual requirements.

National Grid's goal is to establish a long-term working relationship with contractors who share the same safety values and demonstrate those values through their work performance.

Contractor safety at National Grid involves three broad areas:

1. **The Contractor Procurement (Selection) Process**
   Contractor safety begins with the selection of contractors who have demonstrated a strong safety record. National Grid will complete a review during the procurement process that involves determining a contractors' risk and the scope(s) of work involved. National Grid currently uses ISNetworld, Inc. as a third (3rd) party assessment process for assisting with contractor procurement. The 3rd party entity will vet and continually monitor individual contractors' compliance with applicable safety and/or risk and insurance program requirements.

2. **Safety Communication**
   Safety communication covers all the avenues and forums in which National Grid and the contractor communicates safety. Communication begins early in the bidding phase and is on-going as an integral part of the contractor-customer relationship. The goal is to ensure clarity, transparency and to limit misunderstandings.

3. **Safety Compliance**
   Safety compliance is the process of ensuring that the necessary technical provisions of the contract are being followed. National Grid will assign a project representative or other designee to provide guidance and oversight. The Contractor is responsible for their employees and subcontractors and shall be held accountable for ensuring compliance with all applicable safety rules while working on National Grid property, rights of way (ROWs) and our assets.

1.3 Risk Ranking of Work

1. National Grid characterizes and ranks risk by the scopes of work performed. The categories are classified as high, medium or low risk. Prior to being considered for work at National Grid, contractors who perform High or Medium Risk work must be pre-qualified in ISN. See Appendix A for more information regarding the National Grid Contractor Risk Matrix.

2. Activities that are designated as "high risk" means that catastrophic event can result if safety measures are not followed. Activities designated as "medium" risk present certain opportunities for moderate to significant injuries, property or reputational damage, and/or loss of service and/or possibly business continuity. Activities designated as "low" risk may still require safety compliance and control measures, although the contractor performing the work does not necessarily need to be enrolled in ISN, if that is the only classification of work that contractor performs for National Grid.

3. The designation High Risk, Medium Risk, or Low Risk, refers only to the inherent risk associated with the work activity and is not an opinion on the ability of a contractor to work safely.

4. The Procurement Agent will notify the bidder/contractor at the beginning of the procurement process if their contracted service has been ranked as high or medium risk.

1.4 Bidder Information Request – High and Medium Risk Work

1. Any contractor bidding on high or medium risk work shall be required to complete a questionnaire regarding the Contractor's safety program, compliance and history of occupational illnesses and injuries (ISNetworld New Vendor Onboarding application form, located on the ISNetworld website). Contractors will also be required to demonstrate that all employees, including subcontractors, are qualified.

2. ISNetworld then thoroughly reviews contractors' qualifications against a prerequisite list of National Grid criteria. National Grid has established that contractors performing high or medium risk work MUST HAVE and MAINTAIN a grade of "C" or better in the ISN system to perform work and services for National Grid. ISN will track and manage the National Grid pre-qualified contractor bidder lists. This bidder list is the first step for a contractor in establishing a working relationship with National Grid. For active ISN contractors, ISN will request updated information monthly. Contractors who do not have a current PO, contract, or authorized scope of work with National Grid will be considered as a Prospective bidder and will

be asked to submit information quarterly. It is understood that insurance may not be maintained within National Grid standards, however, once a contractor is awarded a contract, proper and adequate insurance must be provided to ISNetworld to achieve a passing grade. Lack of insurance or inadequate insurance is an immediate "F" grade in the ISN system per National Grid criteria.

3. Project representatives may request an exemption or variance from requiring a contractor to be placed in ISNetworld for various reasons. A Supplier Exemption Request form (located in the forms section of the SHE Infonet) shall be completed, documented and signed by the business unit VP and Corporate Safety Director prior to contract award.

4. The information that the Bidder provides National Grid via ISN serves as the basis for assessing safety qualification. For this reason, it is important for contractors to maintain transparency throughout the process. National Grid and ISN will review all submitted information. Any effort in avoiding complete disclosure will disqualify the Bidder from bidding work at National Grid.

1.5 Safety Compliance

1. National Grid representatives evaluate contractor compliance by conducting routine site visits, Compliance Assessments (CA's), Effective Safety Discussion (ESD) visits and attending periodic contractor safety meetings. Contractors should also perform and document safety self-assessments to ensure compliance to federal, state, local and National Grid regulations. This combined effort enhances, solidifies safety compliance and has the added benefit of quality control / quality assurance of the work performed.

2. Contractors bidding on new work shall provide worker qualifications to the National Grid procurement representative via the "Bidder Information Request" form and/or ISN National Grid On-boarding form.

3. If a safety violation is observed by a National Grid representative, the violation will be discussed with the contractor at the time of discovery.

4. Contractor employees enrolled in ISNetworld that are involved in any accident, incident or significant near-miss event, will be required to lead an investigation and root cause determination process. In addition, the contractor must implement corrective actions and establish measures to prevent a recurrence through an incident investigation process and document the details within ISN.

5. Individual contractor personnel who habitually violate any safety rules should be identified, and the contractor should remove the individual(s) from the project. National Grid reserves the right to remove any contractor employee(s) who violate safety rules or procedures; pose a safety risk to themselves, other contractors; our employees; or the general public.

6. If a contractor is observed to be operating in a manner that creates an imminent danger to persons or property, it is the responsibility of all individuals observing the hazard to cease the hazardous operation impacted until the issue has been resolved to the satisfaction of National Grid, the Owners Representative or Safety Representative.

7. Contracts/POs shall require the contractor to immediately forward any citations, notices, or OSHA reportable cases per 29 CFR 1904.39 from a National Grid project, upon receipt to the appropriate project representative and/or ISN. The project representative shall distribute copies of the citation or notice to senior management, Safety, Procurement, and the Legal Department.

8. Willful and/or repeat violations of safety requirements by the contractor may be considered a breach of the contract and reason for contract termination.

9. If the contractor's overall safety performance is viewed as being unsatisfactory or noncompliant with contract provisions, and if the contractor is unwilling to demonstrate satisfactory program improvement, the result may be considered a breach of the contract and reason for contract termination.

10. National Grid project managers and/or construction supervisors shall document safety compliance by completing a "Contractor Performance Evaluation" for each project. This documents both positive and negative safety performance or behaviors and this feedback will be used in the decision process for awarding future contracts.

2.0 GENERAL SAFETY REQUIREMENTS

2.1 Introduction

1. All contractors are required, and expected to comply with all applicable requirements of the Occupational Safety and Health Administration (OSHA), and all other applicable federal, state and local laws, ordinances, regulations, and other project and site-specific permits unless superseded by identified National Grid procedures.

2. This document represents policies and safety-related work methods unique to National Grid and they may be more stringent than OSHA regulations. Contractors must follow these requirements as well as their own rules or regulations that meet or exceed OSHA and other regulatory requirements.

3. National Grid will provide more detailed information and guidance regarding specific procedures prior to commencement of work.

2.2 Applicability

Applies to: All contractors, as needed

1. In any contracted task, where a safety observer is required, it is the responsibility of the contractor to provide that person and ensure that he/she is qualified to perform the role when needed.

2. A 4:1 pitch shall be maintained when using an extension ladder or the ladder shall be tied off and/or secured and 3 points of contact shall be maintained by the climber. If both hands are needed to perform work, then fall protection is required.

3. Although not preferred, if hard hats are worn backwards, the suspension adjuster must always face the rear. Class E hard hats are required for all electrical work.

4. 20KV EH overshoes are required when:
    - Workers on the ground are working within 50’ of the master ground connection point to earth.
    - Operating a wire trailer and pulling/tensioning machine.
    - Operating a winch truck or reel trailer with its payout in an energized area that may result in inadvertent contact.
    - Hand digging in close proximity to energized cables within the tolerance zone.
    - Making repairs in a trench to a faulted primary cable without de-energizing any adjacent energized primary cables within close proximity.
    - Using approved live line tools to move energized primary cables in a trench.
    - If removing underground cable rubber covering or arc suppression blankets from an energized cable.
    - Working within minimum approach distance (MAD) of downed electrical wires or foot patrolling for such wires.
    - If setting poles in proximity to energized lines or equipment and using truck controls from the ground.

5. National Grid expects that all cargo will be secured in accordance with U.S. DOT requirements.

    - As of January 2004, the Federal Motor Carrier Safety Administration (FMCSA) within the U.S. DOT published Cargo Securement Rules 393.100-136 Subpart I – Protection Against Shifting and Falling Cargo.

6. Chaps are required to be worn by ANY person using a chainsaw to make a cut on the ground or assisting in that cut and within striking distance. Other situations where cut off machines are used, chaps designed for the purpose of providing durable protection from abrasion, spatter and sparks from cutting ferrous metals shall be required; however, a hazard assessment should be completed to determine the need. Always use proper cutting techniques and push blades away from the body when using tools that may slip or inadvertently make contact with the leg. Never leave any equipment running while not in use.

7. All applicable contractors must meet the requirements of drug and alcohol testing in accordance with FMCSA DOT 49 CFR Part 40 and Pipeline and Hazardous Materials Safety Administration (PHMSA) DOT 49 CFR Part 199. National Grid shall monitor contractor compliance to the drug and alcohol regulatory requirements through Transportation Advisor or ISNetworld as needed.

8. Contractors who drive regularly in delivery of service for National Grid shall:

    a. Have a safe motor vehicle operations policy which must be communicated to their employees before they begin driving for company business. The contractor is expected to follow National Grid's *Safe Motor Vehicle Operations* policy to include the following:
        - Prior to moving any vehicle, the driver shall perform a "circle of safety" inspection. This is to confirm not any person, animal, equipment, or property will be injured or damaged when the vehicle is moved.

        - Drivers should back into or pull through a parking space so that when you re-enter the vehicle, the first move is forward.

        - No driver shall use a hand-held mobile telephone while driving a vehicle for National Grid business.

        - The driver shall eliminate or minimize sources of potential driving distractions to include, eating, smoking, reading, writing,

grooming, use of any electronic devices, mirror or seat adjustment. These shall be done when the vehicle is not in motion.

b. Comply with all requirements of all federal, state and local regulations regarding safe motor vehicle operations.

c. Ensure that new and existing employees have a valid Driver's License in accordance with requirements of specific job duties and classification/type of the vehicle they are operating. Contractors must have an acceptable driving record. If their driving record is unacceptable, the driver shall not be permitted to operate a vehicle on behalf of National Grid.

d. Provide vehicles in safe operating condition, in accordance with federal state and local regulations. The vehicle should be equipped with proper safety equipment as appropriate for the vehicle type and its intended use.

e. Track and evaluate any vehicular accidents or incidents experienced by their employees. Corrective actions, such as driver coaching, corrective action driver training and medical/vision tests should be recommended and acted upon where appropriate. All accidents or near misses while performing work for National Grid shall be communicated to the National Grid project representative or designee and documented in the ISN system.

f. For more information, contact a National Grid representative for a copy of the National Grid Safety Policy *Safe Motor Vehicle Operations*

9. All contractors that require the use of heavy equipment shall ensure that competent, appropriately licensed, skilled and qualified personnel are in control of this equipment at all times. In addition, contractors shall ensure the following:

- Equipment is inspected for safety and use at the beginning of the work period of shift. All failing or defective equipment and components shall be removed from service.

- Equipment is under the control of trained operators who are always aware of their location and the locations/presence of persons working near the equipment, its swing zones and blind spots.

- Operators shall ensure that no loose clothing or jewelry is present that could inadvertently get caught equipment controls.

- Equipment is kept free of debris, water, oil, grease, mud or anything that could create a slip/fall hazard inside the cab.

- Keep hands, feet, and clothing away from power-driven and moving parts.

- Equipment cab windows should be kept clean and free of mud, ice, snow and/or fog for maximum visibility.

- Never carry passengers on heavy equipment or any equipment unless it is equipped to do so.

- Ensure that stabilizers are extended prior to starting a task.

- Before making a swing, operators shall always look out the windows and mirrors for confirmation that the area is clear. If visual confirmation is impaired or the operator is unsure due to weather, lighting or other interferences, the operator shall cease operation until an independent spotter can check the swing area and confirm it is clear.

- All excavations shall have signs posted, demarcation and controlled to prevent unauthorized persons from entering and falling inadvertently into the excavation. Excavations shall only be opened under the supervision of a competent person for excavation.

- All pot holing/test holing and exploratory excavations shall utilize vacuum excavation whenever near known or the possibility of unknown hazards such as live electrical or gas conveyances. When using vacuum excavation in combination with air blowing/air knife tools, all persons in the immediate area shall be wearing safety glasses in addition to a full face shield.

- No one is to work under a suspended load.

- Never use a bucket to lift personnel.

- Ensure stabilizers are in the upright and stored position before moving equipment.

- Operators shall not leave heavy equipment running unless the following requirements are met:
    - Parking break is engaged and wheels are chocked (if applicable)
    - Surroundings create no hazard of unqualified personnel entering unattended equipment
    - Vehicles and equipment idling limited to that designated state and local environmental regulations (generally, 3 to 5 minutes maximum). See table below for additional information

Region	Vehicles	Idling Limit	Exemptions Include
New York	Diesel trucks	5 minutes	• Traffic conditions • Temperatures < 25°F and motionless for two hrs • Hybrid electric engine charging battery vehicles • To provide power to auxiliary sources
NYC	All Motor vehicles	3 minutes	• Emergency vehicles • Loading/unloading • Temperatures <40°F
New Hampshire	Diesel/ Gas vehicles	5 minutes >32°F 15 minutes -10°F to 32°F No Limit <-10°F and no nuisance created	• Traffic conditions • Emergency vehicles takeoff power for auxiliary uses • Vehicles being serviced or repaired • Operated solely to defrost windshield
Massachusetts	All Motor Vehicles	5 minutes	• Vehicles being serviced or repaired • Vehicles in operation for which associated power is needed • Delivery vehicle in which engine power is needed
Rhode Island	Diesel Motor Vehicles	5 minutes	• Traffic conditions • Operate defrosting, heating, or cooling equipment to ensure health and safety of the driver or passenger. • Temperatures between 0 & 32°F - 15 minutes per hour. If < 0°F idling as needed for heat • To provide power to auxiliary sources • Vehicles being serviced or repaired
Vermont	All Motor Vehicles	5 minutes within any 60-minute period	• Emergency/public safety vehicles while engaged in "official operations" • Idling necessary to operate safety equipment • Vehicles in operation for which associated power is needed • Vehicles being serviced or repaired

- All lifts that occur on National Grid properties, ROWs or near critical assets require formal lifting plans developed by the contractor and reviewed with the National Grid project representative. Some lifts will also require formal critical lifting plans and this may include PE or geotechnical assessments to ensure a stable lifting base for the crane or other apparatus.

10. All temporary, metal fencing installed or located under transmission lines shall be grounded and have signage according to National Grid grounding standards. Contact a National Grid representative for a copy of the Engineering Documents ST 03.05.001ST 03.06.001 and SP 08.00.001.

3.0 ADMINISTRATIVE SAFETY REQUIREMENTS

3.1 Worker Qualification Assurance

1. In order to meet National Grid safety requirements, the contractor must describe how workers, including subcontractors, are qualified. The contractor must supply information concerning the type of skills assessment performed, training programs and how they ensure that employees demonstrate competencies. National Grid reserves the right to review this information and request additional training requirements. All documents shall be uploaded and maintained in the ISN system. For work on process safety assets, the contractor shall ensure all workers and sub-contractors are trained and receive appropriate refresher training to maintain their appropriate level of certification and qualifications needed to perform work safely.

2. The contractor shall provide management personnel qualifications through resumes or other documents. National Grid may interview and/or approve management personnel if considered necessary.

3. For work on Process Safety assets (Gas Transmission, Generation, LNG, LNG Transportation and CNG), contractors shall provide a description of their experience in the business asset and specific tasks including similar projects, lists of licenses/certifications, and references from previous similar projects. Contractors shall be made knowledgeable of National Grid process safety requirements that are relevant to their specific work activities by the business hiring them.

3.2 Meetings

Applies to: All contractors; as needed

1. The pre-bid meeting is coordinated by National Grid Procurement to provide bidders with an opportunity to become acquainted with contractual requirements and specific safety issues concerning the project, including company-specific safety rules and known site conditions.
   a. For contractors working on Major Hazard Assets, contractual

language including designation of site medical facilities, locker rooms, bathrooms, etc. should be discussed by the project team with the contractor at this time.

2. At this time, Procurement will notify the prospective bidders of the following:

   a. If they are required to submit a project safety plan (HASP) prior to the pre-construction meeting

   b. The cost of specific safety equipment, practices and personal protective equipment shall be factored into their bid/proposal.

3.3 Project Health & Safety Plan (HASP)

Applies to: Contractors performing high or medium risk work

All HASPs shall be submitted to the National Grid project representative for review and approval before work commences. The National Grid project representative shall ensure the HASP meets National Grid criteria and includes all aspects of the project prior to a review by Safety.

1. Contractors who perform high or medium risk-ranked services shall submit a project-specific HASP plan prior to the start of the project and/or at pre-construction meeting. Your National Grid project representative will provide you with specific requirements of the format and/or forms to be completed.

2. For typical work that can be completed in 1-2 days, a documented job brief is sufficient to address the hazards and proper mitigations. Ask your National Grid project representative for details.

3. The *Short Version* plan shall be used for work that generally last 3-5 days. This plan shall include a completed safety hazards checklist and the Emergency Contact Sheet. Proper mitigation shall be documented in the safety hazards checklist for any applicable hazards. For examples of the Safety Hazards Checklist and the Emergency Contact Sheet, a National Grid representative can provide related forms under the *Contractor Safety* website in the Infonet.

4. The *Long Version* plan shall be used for work that will take at least 6 days or complicated projects. At a minimum, the *Long Version* safety plan shall include the following elements:

   a. Roles and Responsibilities

   b. Scope of Work

c. For contractors working on Major Hazard Assets - List of all equipment contractor is expected to use in work activities and indication that it meets regulatory and National Grid requirements

d. For contractors working on Major Hazard Assets - List of contractor materials to be brought onto work site for review and approval by National Grid

e. Task and hazard identification and risk assessment of the hazards

f. Hazard mitigation/control procedures and work methods

g. Incident investigation and reporting

h. Compliance and monitoring

For an example of a Long Version HASP, a National Grid representative can provide related policies and procedures under the *Contractor Safety* website in the Infonet.

5. A site-specific HASP that meets the requirements of 29 CFR 1910.120(b)(4)(ii) must be prepared by the contractor for all work at contaminated sites. The HASP must include at a minimum:

a. A safety and health risk or hazard analysis for each site task and operation

b. Personal Protective Equipment to be used by employees for each of the site tasks and operations

c. Medical surveillance requirements

d. Frequency and types of air monitoring and personnel monitoring to be used

e. Site control measures

f. Decontamination procedures

g. An emergency response plan for safe and effective responses to emergencies, including the necessary PPE and other equipment

The contractor/National Grid project representative shall contact the Environmental Department for guidance on site requirements and to initiate any required regulatory notifications.

For contractors performing bulk commodity transportation activities, a risk assessment including the potential consequences, frequency and safeguards to be used shall be performed and included in the HASP. If a preexisting National Grid requirement is in place for managing bulk commodity transportation activities, one shall follow those requirements, with no additional risk assessment being required.

Every contracted and subcontracted employee performing work on the project shall review the HASP to ensure steps in the plan are adhered to mitigate hazards during the pre-job orientation. These mitigation steps shall be incorporated into daily job briefs.

In addition, all workers shall sign an attendance sheet during the pre-job orientation that they have reviewed the plan, will adhere to the mitigation steps and they fully understand the plan. This document will be kept at the job site and available for review as needed and if requested by any National Grid representative, or any other parties.

A. Roles and Responsibilities

The HASP shall identify who is providing project oversight and how they are qualified. For example, if the work requires excavation, there must be someone on-site who is qualified as an excavation competent person.

For multi-employer work-sites, the general contractor is responsible for all their employees and subcontractors. The safety plan shall clearly state this responsibility.

If requested to do so, Contractors shall designate a competent person to participate in or conduct a process hazard analysis (PHA) regarding a portion or the entirety of the project. National Grid will not be responsible for training the contractor on the PHA methodology.

B. Scope of Work

The Contractor shall clearly and briefly state the scope of work as provided by National Grid. The plan must specifically address the project or services requested by National Grid.

C. Task and Hazard Identification and Risk Assessment

The contractor shall perform a risk assessment by identifying all significant tasks, the anticipated hazards and hazard mitigation procedures.

The contractor's cost to provide adequate safety measures and to comply with National Grid requirements must be considered and budgeted in the bid/proposal.

D. Hazard Mitigation Procedures and Work Methods

For each hazard, the contractor shall specify measures that will be taken to eliminate, control or mitigate these hazards.

A table below is an example of a method to simply and clearly organize and present the task, hazard, and mitigation steps:

Location: Substation Yard		
Task	**Hazard**	**Mitigation Steps**
Material Handling	Contact with overhead energized lines/equipment	Off load in the clear and have a safety observer present

E. Incident Investigation and Reporting

All work related incidents involving injury or illness to employees, the public or property damage (including contractor vehicle accidents) shall be reported to the National Grid project representative and documented in the ISN system.

F. Compliance Monitoring

To ensure that both contractor employees and subcontractors will achieve safety compliance, jobs with over 100 workers at any point in time or in excess of $1 million will require a full time safety professional hired by the contractor. This safety professional must be qualified, competent and be on site anytime work is performed. Qualifications of this safety representative must be acceptable to National Grid prior to hire by the contractor and may be documented in the ISN system.

G. Environmental Compliance

Unless otherwise specified and based on the scope of work, any potential environmental risks shall be determined and addressed by the

contractor following all applicable National Grid procedures. For more information, contact a National Grid representative regarding Environmental Procedure No.6 *Contracted Services* and Environmental Procedure No.25 Appendix A, *Environmental Screening Checklist.*

3.4 Contractor Orientation/Pre-Construction Meeting

Applies to: All contractors, as needed

1. A National Grid project representative, construction supervisor, or other designated National Grid representative may hold a contractor orientation or pre-construction meeting prior to the start of a project/service. Other attendees may include; the Safety department, Environmental representatives, as well as contractor management as needed.

2. It is intended to serve as a method to provide the contractor with the tools necessary to educate their employees and subcontractors on National Grid's procedures and requirements. The session is not intended to train the contractor management, their employees or subcontractors.

3. All contractors are required to attend a National Grid orientation program specific to the type of work they will be performing. Contractor management representation shall also be present meeting and all documentation of attendance shall be kept at the job site and available to any National Grid representative. For visitors and contractors working on Major Hazard Assets, site orientation shall at a minimum include the following:

   - General site hazards
   - Specific hazards involved in each task the employee may perform
   - Safety policies and work rules, including Process Safety policies
   - Location of emergency equipment like fire extinguishers, eyewash stations, and first-aid supplies
   - Smoking regulations and designated smoking areas if applicable
   - Steps to take following an accident or injury
   - Proper reporting of emergencies, accidents, and near misses
   - Selection, use, and care of personal protective equipment
   - Emergency evacuation procedures, routes, and security systems
   - Safe housekeeping rules

- Safe use of tools and equipment
- Hazardous materials in use and location of safety data sheets

Site access shall not be granted to contract employees working on process safety assets until orientation is conducted.

4. The contractor's Project Health & Safety Plan will be discussed at this meeting including a final review of the safety hazards checklist to ensure proper hazard identification and mitigation plan has been implemented.

5. These hazard mitigation measures shall be reviewed and work shall not commence until these hazards have been adequately addressed. The National Grid project representative will discuss the methods by which compliance will be achieved to National Grid safety requirements with the contractor.

6. An Emergency Call List shall be exchanged with the National Grid project representative for high or medium risk projects or as applicable. This list must contain 24-hour contact information for key contractor and project personnel, including the project representative and Safety representatives. This list should be distributed to all concerned, as determined by the project team, prior to the start of work. For contractors working on process safety assets who have an emergency response role, the emergency response plan shall be updated to clarify the contractor's role in the event of an emergency on site.

7. For routine maintenance services, a review of associated safety issues and specific facility issues, restrictions or practices, such as evacuation procedures, shall be discussed with the contractor upon initial hiring. Any changes in the facility that may affect the safety of contractor or National Grid employees or third parties must be communicated immediately.

8. Upon completion of the contractor orientation or preconstruction meeting, the contractor management official shall certify in writing that: (1) the contractor has been informed of National Grid safety requirements; (2) that employees and subcontractors have the appropriate qualifications to perform the work, and; (3) the contractor agrees to comply with all applicable safety requirements. The certification shall be completed annually in ISNetworld as an acknowledgement of the above requirements.

3.5 Job Safety Briefs

Applies to: All contractors; as needed

1. Job safety briefs shall be documented in writing. Written job safety briefs, permits, and/or plans shall be available at the job site for inspection and retained for 30 days after the job is completed.

2. National Grid reserves the right to perform a safety stand-down with any contractor for purposes including, but not limited to: recent injuries, incidents or near misses; identified hazards at job site or equivalent, and for other reasons to communicate with the contractor crew.

3. Each crew shall conduct these job safety briefs prior to each day's work at the job location, when there are changes to the day's work order or plan, and when a new worker or company joins the crew additional briefs are required.

4. Each worker must have the opportunity to voice concern. The work cannot begin until each worker signs off on the job safety brief stating that they have discussed the work, raised any questions, and agree with the plan.

5. Visitors to the work site shall be asked to read and sign the job brief acknowledging they understand contents. Contractors shall review the job brief and discuss the elements of the hazards and mitigation steps with each visitor prior to entering the job site. If a visitor refuses to sign, the general foreman will note it on the brief and will not allow the visitor to enter.

6. SITE SIGNAGE: An assessment of the work site should be conducted by the National Grid project representative overseeing the work with the contractor to determine if site signage will be needed to protect site visitors, the public or any other persons entering the work site. If Site Signage is required at the site, the signage shall be posted at the main entrance to the work site. The sign shall direct all visitors to check in with the Person in Charge (PIC), be escorted to the designated safe area and advised of all work currently in progress. The visitor is expected to comply with all related safety requirements and sign off on the Job Brief before entering the work site.

3.6 Safety Meetings

1. In addition to job safety briefs, the contractor shall have regular safety meetings with their employees and subcontractors. Contractors performing high or medium risk work shall have weekly safety meetings, while low risk contractors, at a minimum, shall have safety meetings monthly and attendance must be documented.

2. The safety meetings shall include the following topics: statistics, incidents, near misses, updates on old business and new business raised. It will include the round table discussion by the workers and the action items discussed. Meeting minutes must be documented and shall include specific action items, their due dates, persons responsible and a completion date. This documentation shall be available for inspection during the project period, and for 30 days after the project is completed. For contractors working on Process Safety assets, meeting minutes from contractor shall be shared and discussed with National Grid site management.

3. Routine Safety meetings/calls between National Grid and the Contractor shall be coordinated on a regular basis. Safety meetings may include but are not limited to ESD/Compliance Assessments, Safety Briefs, Safety Day discussions and regularly scheduled calls to promote safety and best safety practices. Contractors working on Process Safety assets for more than 6 months shall schedule leadership visits to discuss process safety topics.

4. Contractors working on process safety assets for greater than three (3) months, or as needed, shall hold project planning meetings to discuss short term and long term work items. Project planning meetings shall include safety performance monitoring against project targets and should include a National Grid SHE representative for jobs on Major Hazard Assets in addition to a National Grid site representative.

3.7 Incident Investigation

Applies to: All contractors (regardless of risk ranking)

1. All contractors are required to report any work-related incidents involving injury or illness to employees, the public or property damage to the National Grid project representative. The first priority is to ensure that anyone injured receives medical treatment. Examples of incidents may include, but is not limited to: injury, property damage, adverse public impact, near miss, a hazardous condition and road traffic collisions (RTC).

2. Contractors will then be responsible to perform an incident investigation immediately following the incident and document root cause/corrective actions in the ISN system and to National Grid.

Incident Response Steps

In the event of an incident, the contractor shall provide details of the incident to National Grid that follows the steps below.

1. Contractor supervisor collects basic information about the incident from the employee or witnesses:
    - What happened?
    - Who and how many people were injured?
    - What treatment was administered?
    - What was the nature and seriousness of the injury?
    - Where did the incident occur?
    - When did the incident occur (date, time of day)?
    - Were there any witnesses?

2. Contractor supervisor immediately calls the project representative or other National Grid point of contact. All incidents shall be entered into the Incident Management System (IMS) as soon as possible by the National Grid project representative or National Grid designee. When dialing 1-866-322-5594, the caller will be prompted to select option 2 for anything other than an employee injury.

3. Contractor shall conduct an investigation within 24 hours of the incident that will identify contributing factors and root cause analysis relating to the incident and the corrective actions that will be taken to prevent future occurrence. This information will be documented in the ISN system.

4. Contractor vehicle accidents occurring during the performance of work will also be investigated and reported to National Grid.

Other Reporting

National Grid may periodically request the following annualized data for all work activities limited to National Grid operations:

- Lost Time Incident (LTI) rate for workers
- Restricted Work rate
- OSHA Recordable Incident (ORI) rate

4.0 TECHNICAL SAFETY REQUIREMENTS

4.1 Personal Protective Equipment (PPE) Requirements – General

Applies to: All contractors (regardless of risk ranking)

1. Basic PPE attire at construction sites and other similar work zones include, at a minimum: hard hat, safety shoes and safety glasses with side shields, and ANSI certified Class 2 or Class 3 Hi-visibility (hi-vis) clothing / vests. Contractors shall refer to the US Department of Transportation's Manual on

Uniform Traffic Control Devices (MUTCD) to determine the correct class of hi-vis clothing / vests or wear the higher class. The contractor and their employees, including subcontractors are expected to follow the same rules and protocols as National Grid personnel.

2. The contractor shall ensure that their employees and subcontractors use protective safety toe footwear when working in areas where there is a danger of foot injuries due to falling or rolling objects, or objects piercing the sole, and where such employee's feet are exposed to electrical hazards. In addition, during inclement weather conditions or adverse events, the addition of anti-slip footwear or outer foot wear may be appropriate. Electrical Hazard (EH) rated footwear is required at all times. EH rated overshoes are required when working on or near (within Minimum Approach Distance MAD) around electrical equipment over 50 volts, when working in electric substations, excavating in and around electrical duct banks or in an area of expected downed wires per OSHA 1910.136 and ASTM standard F-2413-05.

3. Guidance for additional PPE is referenced in other sections of this document.

4.2 Flame Resistant Clothing Requirements

Applies to: All contractors; as needed

1. Flame Resistant (FR) clothing shall be worn when personnel work on energized equipment/lines or when distance and position will expose the worker to electric arc or flame hazards. FR clothing shall also be worn during live gas work as outlined in the gas PPE Matrix (Gas Policy SHE01001) and within LNG operations locations as required. FR clothing also includes arc- resistant rain gear. This additional ensemble may also be required as part of the job for contractor personnel. Contact a National Grid representative for a copy or to view the PPE matrix.

2. FR clothing shall be worn as the outermost layer of clothing and when workers measure voltages, test or ground electrical equipment/lines.

3. FR clothing shall be worn when work requires the use of rubber protective equipment or the use of insulated live line tools.

4. FR clothing shall be worn when workers control/operate electrical equipment over 50 volts at the device location or are within 10 feet of equipment which is being physically operated/ worked on by another worker.

5. Visitors are not required to wear FR clothing in substations or production plants unless they are engaged in electrical work. The National Grid project representative will be able to determine whether FR clothing will be required based on the specific contractor task. Note: Gas contractor FR requirements may differ slightly. Please refer to National Grid PPE Matrix for Gas operations within Gas Policy SHE01001 as needed.

6. FR clothing shall meet a minimum arc rating of 8 cal/cm^2 (HRC 2) for energized electrical equipment unless otherwise specified based on increased potential exposure as indicated in the Arc Flash Tables in H-807 *Arc Flash Analysis and Mitigation* program.

7. Additional FR clothing protection is required when performing work on the distribution system in NY North and New England (legacy National Grid) stations. Contact a National Grid representative for a copy. (NG Employees: If the link does not work, copy and paste the url into your internet browser)
http://us3infonet/sites/eng_delivery_svcs/Pages/ArcFlashMitigation.aspx#2015

8. Contractors who may be involved with tasks requiring the implementation of this program shall be informed by National Grid. Contractors will be required to follow all aspects of OSHA and any other applicable regulation as it applies to the tasks they perform.

4.3 Rubber Gloves and Sleeves

1. Rubber glove use is required for work on all electrical apparatus at 50 Volts or greater. Rubber gloves shall be donned before the worker leaves the ground and shall be worn until the worker returns to the ground (commonly referred to as “ground to ground”, “cradle to cradle”)
2. Class 0 gloves are required for exposures up to 1000 Volts.
3. Class 2 gloves are required for voltages between 1000-15,000 Volts.
4. Rubber sleeves must be worn where work is conducted within the MAD of primary electrical apparatus that is not tested, de-energized and grounded.
5. For voltages 23 kV and above, workers can use specialized equipment or work practices as long as these workers have been appropriately trained and qualified. National Grid may request training records from the contractor.
6. Rubber glove exceptions for specific jobs (other than those listed in this section) are permitted only with the dated, written approval of a Division Director.

7. It is the contractor's responsibility to wear class 2 rubber gloves when grounding trucks or equipment due to a possible difference in potential.

Exceptions

No rubber gloves are required:

- When working in a properly established equi-potential zone.
- When the operator remains at the same potential as the equipment by being off the ground and on the equipment.
- When a qualified worker performs transmission "hot stick" work on lines 69 kV or greater and no other energized wires are on the pole or structure below the worker.
- When work is performed on transmission structures carrying only energized conductors (115kV and above) and the Live Line Techniques are not being employed. While performing these activities, the worker shall utilize conductive clothing such as conductive gloves, boots, leg straps and/or any other applicable conductive clothing.
- When climbing a steel structure to perform structural reinforcements while maintaining MAD from energized conductors or apparatus.
- When climbing a steel structure to access an area that has been properly grounded.

4.4 Isolation of Energized Apparatus

1. **Non-Reclosing Criteria and Live-Line Maintenance and Construction:** The appropriate interrupting devices (breakers, reclosers, circuit switches, etc.) will be placed on NON-RECLOSING in accordance with National Grid tagging procedures.

2. **Tagging Out Lines or Apparatus**

   The National Grid Construction Supervisor or designee shall coordinate all switching and tagging in accordance with the most current EOP on Clearance and Control.

Upon receipt of Clearance, the project representative will present the Contractor's Person in Charge with the "Contractor Permission to Work Form" (Form NG0060), which states the specific apparatus that has been de-energized and that certain device(s) are tagged in the Protective Position and will remain so until the Contractor's Person in Charge informs the construction supervisor or designee of the completion of the work utilizing the "Contractor Completion of Work" section of the "Contractor Permission to Work" form.

The original transferred copy needs to be returned after the completion of work section is filled out & signed. In some cases the tailboard is outside & is susceptible to elements & damage; a copy shall be utilized in the field instead of the original.

No work will be performed until the "Contractor Permission to Work Form" is received from the construction supervisor stating that the equipment has been de-energized and a clearance to work has been given. The Contractors Permission to Work Form and a written grounding plan shall be attached to the crews Job Briefing and be kept at the work location.

After the "clearance" is received from the National Grid Construction Supervisor, the various substation conductor bus and equipment to be worked will be tested and "Grounds" installed. Grounds shall be rated for the fault current of the line/equipment being grounded. (Note: Rubber Gloves and FR clothing are required when installing and removing grounds). The contractors "Person in Charge" (Construction Supervisor/General Foreman) shall be responsible for determining the location and number of grounds.

Vehicles and equipment may utilize a single 4/0 cu for grounding inside the substation. Employees working on de-energized lines and equipment shall always work between grounds.

Prior to the application of any personal protective grounds, the circuit to be worked on must be tested for the presence of voltage using an approved potential detector. The worker must verify the detector is in operating order prior to and after testing for voltage. MAD must be maintained during the testing, and appropriate PPE shall be worn. Testing for voltage shall be done at the point where the grounding devices are to be attached. All phases of the circuit to be worked on shall be tested at each location that grounds are installed.

When an Air Gap is required to create a work zone, the component (a tap) shall be removed in whole from the system. When National Grid switches

out lines or apparatus, any grounds that may be installed shall only be considered a visual reference, and shall not be considered a means to protect the Contractor's employees. The Contractor is responsible to install their personal grounds, in accordance with all OSHA, Federal, State and local safety procedures. National Grid may provide guidance on the minimum size of the grounds to be used based on circuit available fault current. Refer to *Electric Operating Procedure D002,* for applicable grounding size. Ground rods shall be fully driven into the earth away from the workers and work area. T-Bar ground rods are not to be used on National Grid property.

The National Grid Construction Supervisor shall review the contractor's plan for the quantity and locations of grounds, ensuring that the work the contractor is performing is between grounds, covering all potential sources. All three phases shall be grounded. (In stations, from each phase to the station ground grid). Grounds shall be placed as close to the work area as reasonably possible, between the work area and all potential sources of inadvertent energization. A copy of the grounding plan shall be kept with the job safety brief.

It is the contractor's responsibility to account for all their grounds. The contractor shall provide, maintain, and enforce a ground tracking program suitable to National Grid. In the instance of a zone expanding/collapsing, remaining grounds shall be listed on the Contractor Permission to Work Form and verbally communicated to the construction supervisor.

3. Grounding Mobile Equipment

When mobile equipment requires grounding, it shall be solidly grounded by means of appropriate sized copper cable while using rubber gloves. The cable shall be fastened to a securely attached clean metallic portion of the equipment, or shall be fastened to a grounding stud provided for the purpose at one end and an adequate ground at the other end.

Non insulated booms such as digger derricks that have the possibility of encroaching the MAD shall be grounded and barricaded. The ground is to trip the circuit and the barricade is to protect anyone who may become in contact with the truck during this energization.

4. Minimum Approach Distance (MAD)

Refer to OSHA 29 CFR 1910.269 for more information and details regarding qualified and unqualified workers.

4.5 Appointment of a Safety Observer

A safety observer shall be required if an employee (operator) determines that it is difficult to accurately determine the distance between the equipment (minimum approach distance) and energized parts. The Safety Observer shall never be a substitute for minimum approach distance (MAD), personal protective equipment (PPE), insulate/isolate techniques or work area identification as a form of employee protection.

The person in charge of the work (contractor or National Grid), shall appoint a qualified employee or employees to perform the task of a safety observer. The personal in charge shall:

1. Ensure a documented job brief is completed and includes the name of the safety observer, additional subjects such as the location of gas lines, energized equipment, in or adjacent to the work area and the limits of any de-energized work area
2. Discuss the scope of work and communication techniques used to warn or notify the equipment operator of hazardous conditions.
3. Communicate any changes to work and job completion to the safety observer
4. Select another safety observer if there is a need for the existing observer to have break in service.

The safety observer is a qualified employee who has been appointed by the person in charge based on the hazard assessment and the job brief. The safety observer shall:

1. Observe the worker performing the task/activity until all hazards have been eliminated or the task/activity has been completed
2. Have shown proficiency in the task/activity being observed and have a full understanding of the job and the hazards associated with the task/activity.
3. Remain continuously focused on the task/activity being performed and not perform or assist any other job activities while observing the worker performing the task/activity
4. Notify the person in charge if there is a need to have a break in service. Work must stop until a new observer is appointed or the safety observer returns.

A safety observer shall also be required when a critical lift is being performed. A critical lift plan shall be required during the following circumstances:

1. An object is lifted over energized apparatuses or assets where a failure of the lifting equipment or rigging could result in a significant safety hazard or cause significant disruption in service to National Grid customers.
2. The crane or other lifting apparatus is anticipated to be operated above 80% of its rated capacity for the specific load chart for the lift.
3. A man basket (pinned or suspended) is to be utilized. All fall protection rules shall be followed when in a man basket.
4. Two cranes will be used in concert to lift a single object
5. Internal substation construction involving all power transformers, control houses, capacitor banks and transmission breakers.
6. Lifts in LNG or Gas plants where a hazard assessment or job brief identifies a significant risk.
7. The lifted load will be less than twice the minimum approach distance (MAD) of the nearest energized part. Until a qualified electrical worker confirms the MAD, loads and equipment shall maintain a 20 foot distance. Once nominal voltage is established, the MAD will be according to OSHA tables.
8. The lifted load is hoisted over buildings or the general public.

4.6 Work Zone Traffic Control

1. If work activity is on or near a road, the contractor and their subcontractors shall comply with all applicable parts of the most current US Department of Transportation's Manual on Uniform Traffic Control Devices (MUTCD), state, local Work Zone Traffic Control requirements and the National Grid Work Zone Traffic Control Manual. Please contact your National Grid representative for a copy of the manual found in the Safety Homepage on the Infonet.

2. If pedestrian traffic is disrupted, pedestrians should be provided with a path that is reasonably safe, convenient and accessible. Pedestrians should not be led into conflicts with work site vehicles, equipment or operations.

3. If working in areas covered by state permits issued to National Grid, contractors shall comply with the provisions (work practices and notifications) of the permit language. These permits must be available on the job site upon request.

4.7 Qualified Gas Worker

Applies to gas projects/activities

1. Gas contractor employees will be operator qualified as required and defined according to the Code of Federal Regulations, Transportation, 49, Subpart 192.801 through 192.809.

2. Until these qualified employees have demonstrated proficiency in the work practices involved, they are considered employees undergoing on-the-job training and must be under the direct supervision of a qualified person at all times. According to the definition of a "qualified employee", the employee also must have demonstrated an ability to perform work safely at his or her level of training.

3. National Grid requires contractors with gas qualified employees to provide documentation on how they qualify their workers.

4. Additionally any qualifications' of contractor personnel shall be in full accordance with the Company's Operator Qualification written plan, (OQ Plan) Refer to the most current list of covered tasks in accordance with National Grids' Operator Qualification Program and the Northeast Gas Association, (NGA).

4.8 Qualified Electrical Worker

Applies to electrical projects/activities

1. According to 1910.269(a)(2)(ii), a qualified electrical employee must be trained and competent in the following prior to starting work:

    - The skills and techniques necessary to distinguish exposed live parts of electrical equipment
    - The skills and techniques necessary to determine the nominal voltage of exposed live parts
    - The MAD specified in 1910.269 corresponding to the voltages to which the qualified employee will be exposed
    - The proper use of special precautionary techniques, personal protective equipment, insulating and shielding materials, and insulated tools for working on or near exposed energized parts of electrical equipment

2. Until these qualified employees have demonstrated proficiency in the work practices involved, they are considered employees undergoing on-the-job training and must be under the direct supervision of a qualified person at all times. According to the definition of a "qualified employee", the employee also must have demonstrated an ability to perform work safely at his or her level of training.

3. National Grid requires contractors with electrically qualified employees to provide documentation on how they qualify their workers.

4.9 Qualifying Non-Electrical Worker

Applies to: All qualifying non-electrical contractors working near energized lines and equipment; as needed

1. The contractor shall provide orientation for non-electrical workers entering and working within restricted areas such as a substation and those working near energized lines and equipment.

2. The information provided to these workers must meet the requirements of paragraph 1910.269(a)(2)(ii). However, the orientation and training may not be as comprehensive as the qualified electrical worker would be.

   They must know:

   - What is safe and not safe to touch in the specific areas they will be entering;
   - The maximum voltage of the area;
   - The MAD for the maximum voltage within the area;
   - Proper use of personal protective equipment and in the work practices necessary for performing their specific work assignments within the area.

3. Until these workers have demonstrated proficiency in the work practices involved, they are considered to be employees undergoing on-the-job training and must be under the direct supervision of a qualified person at all times.

4.10 Asbestos, Lead and other Hazardous Materials

1. Asbestos and lead materials associated with electrical and gas equipment includes, but is not limited to: cement-type cable covering, cable wrap, wire

coatings, coal tar pipe wrap, and transite panels and conduits. Asbestos and lead materials may also be present in building materials including but not limited to: paint, mastics, caulking, insulation and roofing materials.

2. Where asbestos and other hazardous material is present and likely to be disturbed, the National Grid project representative and contractor shall coordinate how the asbestos, lead or other hazardous materials will be managed and shall consult National Grid's Safety & Environmental department as appropriate.

3. Removal of this material must be done by individuals specifically trained and qualified to handle asbestos and lead. Refer to National Grid Safety Procedures, F-615, F-617 and F-619 for guidance on asbestos and lead handling and removals. For more information, contact a National Grid representative for a copy of these procedures.

Note: Contractors who will encounter asbestos or lead as part of their work shall reference in their safety plan how they will address this hazard.

4.11 Lift Plans for Work Near Energized Electrical Equipment

1. All work involving hoists, cranes or other lifting equipment **within 10' of energized electrical equipment** must have a detailed lift plan/procedure.

2. As a minimum Lift Plans shall include the following:

    a. Designated Operator and Signal person

    b. Detailed travel and flight path that ensures the boom and material being raised is controlled 100% of the time and will maintain the appropriate clearance

    c. Designated cover up and isolation to ensure employee and equipment safety in the event of an unplanned action or failure

    d. Emergency action plan with detailed instructions to respond to unplanned/uncontrolled event during the lift or positioning of the lifting equipment.

    e. Documented load weight and equipment lifting limits

f. Rigging equipment and methods that will be used during the lifting. Sign off/approval from the management official responsible for the work

4.12 Fall Protection

1. Fall protection or fall restriction devices shall be used when working at heights over 4 feet with the exception of ladder use, when three points of contact are maintained. Examples of fall protection include appropriate guardrail systems, completed/approved scaffolding and personal fall arrest. For more information, see the National Grid Safety Procedure H806 *Fall Protection*.

2. All fall protection shall be inspected before use each day to determine if equipment is in good working condition. Defective equipment shall not be used and shall be removed from service.

3. A worker may enter or exit an aerial lift (at heights above four (4) feet) provided that fall protection such as guardrails or a fall arrest system is used while the worker moves between the lift and the working surface. Before any such transfer is made, the employee shall be properly tied-off to an adequate support, the pole or structure prior to and in the direction of the transfer.

Exceptions to fall protection shall be in accordance with Federal & State requirements.

4.13 Herbicide Application

1. Vegetation spraying shall be conducted unescorted only by contractor employees who have been designated as a Qualified Electrical Worker, where applicable.

2. The spray applicator shall have ID cards issued by Security with background checks available from the contractor.

3. National Grid management shall require a schedule of the spraying in their areas.

4. Once spraying begins, the contractor must contact local management on a daily basis to inform them of progress or changes to the schedule.

5. The contractor shall post all stations with dated signs indicating when the station was sprayed. These signs should not inhibit access to the station.

6. The contractor shall ensure that any stored materials and equipment do not get covered with “overspray”. Overspray represents a substantial safety hazard and cannot be allowed.

7. When applying herbicides, contractor employees shall wear appropriate PPE in accordance with product labels.

5.0 UNDERGROUND OPERATIONS WORK

In addition to the other requirements referenced in this document, this section covers requirements that are specific to underground operations work.

5.1 PPE Requirements

All contractors shall comply with the applicable PPE and WZTC requirements referenced in Section 4.0.

5.2 Enclosed Space Assessment, Ventilation, Entry and Rescue

Refer to the National Grid EOP-UG006 *Underground Inspection and Maintenance* and National Grid Safety Procedure I-902 *Enclosed Space Procedure* for more information regarding enclosed space requirements.

1. Contractors are required to follow all procedures in this document in regards to enclosed spaces (manholes, sidewalk vaults, etc.), including assessment, ventilation, entry and rescue.

2. Each enclosed space shall be tested prior to removing manhole lids and entry. Atmospheric testing shall be continuous for the duration of the entry using a calibrated, industry approved atmospheric tester.

3. When performing hot lead work or when indicated by atmospheric monitoring, engineering controls such as forced mechanical ventilation shall be used when working in National Grid manholes at all times.

4. All contractors who are qualified electrical workers will treat these spaces as “enclosed spaces” and follow non-entry rescue provisions.

5. In some situations a boom is allowed for retraction from an enclosed space. Refer to Safety Procedure I-902 for more information.

6. Steel cable or wire rope for non-entry rescue is prohibited.

5.3 Equipment Safety Inspection

1. Inspect underground facilities (manholes, vaults, hand holes, splice boxes, junction boxes, pad mount transformers, switchgear and submersible equipment, etc.) each time a crew performs work at one of these facilities. All separable components in these facilities shall be inspected by infrared instrumentation. A National Grid representative can provide details from the National Grid EOP-UG001 *Infrared – Non-Contact Thermometer Inspection Requirement for Underground Equipment* for more information.

2. The infrared (IR) thermometer or camera shall, at a minimum, have a range of -25°F to 1400°F with a plus or minus 1% accuracy. For more details and current operating procedures, contact a National Grid representative regarding EOP UG001.

3. The format for data collected shall follow the National Grid EOP UG006 *Underground Inspection and Maintenance* requirements. Please contact a National Grid representative for more information.

4. "Touch Potential" testing of metal street lighting poles is required as a part of any maintenance work. For more information, a National Grid representative can provide a copy of the National Grid EOP G016 *Elevated Equipment Voltage Testing* and National Grid Work Methods Bulletin #04-26 *Touch Potential Testing of Metal Street Lighting Poles.*

5. Touch Potential testing results shall be recorded on the job safety brief and the manhole inspection form which shall be given to the National Grid Construction Supervisor or designee.

6. All contractors working for National Grid shall use materials and equipment in accordance with the manufacturing guidelines. It is the contractors' responsibility to understand the manufacturers' limits and prescribed use of their tools and equipment before each use.

7. Workers shall test and verify that the underground cable is de-energized and guillotine the cable if needed from outside the hole. Rubber gloves shall be worn at all times while performing this task.

6.0 OVERHEAD LINE WORK

In addition to the other requirements referenced in this document, this section covers requirements that are specific to overhead line work.

6.1 PPE Requirements

All contractors shall comply with the applicable PPE and WZTC requirements referenced in Section 4.0. In addition, contractors will follow ground-to-ground and cradle-to-cradle use of rubber gloves while performing work on energized overhead lines. Any foreign wire, including those on a pole or structure constitutes an energized source and requires the use of rubber gloves (ex: Cable TV, telephone, fire alarm wire, etc.).

6.2 Fall Protection

All contractors who climb structures such as wood poles or transmission towers shall utilize enhanced fall protection equipment (fall arrest devices) and techniques (ex: *Buckingham Buck-Squeeze, Miller StopFall* or *Jelco Pole Choker*). When working on wooden and steel structures, a full body harness and lanyard shall provide 100% fall protection at all times (100% tie off, Shepperd's Hook, etc.). Climbers shall never be allowed to drop or slide down a pole or structure more than two feet.

Exceptions to fall protection shall be in accordance with Federal & State requirements.

6.3 Pole/Structure Inspection

Contractor shall ascertain the structural integrity of the pole or other structure prior to installation, removal, repair or modification of the equipment on the structure.

1. Prior to climbing any pole, an inspection and test of the condition of any pole being climbed shall be performed. The weight of the employee, the equipment being installed and other working stresses (such as the removal or re-tensioning of conductors) can lead to the failure of a defective pole or one that is not designed to handle the additional stresses.

2. If the pole is found to be unsafe to climb or to work from, it must be secured so that it does not fail while an employee is on it. The pole can be secured by a line truck boom, by ropes or guys, or by lashing a new pole alongside it. [29 CFR 1910.269(q)] If measures cannot secure the pole, the contractor must cease operations and notify the National Grid Construction Supervisor or designee

6.4 Electrical Work Methods

1. Jumpers of any type shall not be used to keep transformers, risers or transformer banks energized for the purpose of changing potted porcelain cutouts. A National Grid representative can provide information to the National Grid Electric Operation Procedure (EOP) D001 *Cutouts – Open Type* for more information.

2. Potted porcelain cutouts must be changed out when work is being completed on a pole even if this is not planned in the scope of the work provided.

3. Properly rated and inspected slings, chains or tongs shall be utilized to move poles and equipment. Winch lines must not be wrapped around poles or looped around transformer ears to lift without a sling or chain.

6.5 Transmission Overhead Lines

1. For work on transmission circuits, red tape shall be placed around any energized pole, pole structure, or tower adjacent to the de-energized line.

2. When one circuit of a double circuit pole or tower line is de-energized for work, a red or orange flag shall be placed on the energized side of the pole or tower nine feet below the lowest energized conductor. In addition, a red or orange flag shall be placed on the lower cage on the side toward the energized circuit at each arm level as employees work on them or pass them.

3. All contractors using ATV's, UTV's or RTV's for transmission or forestry work, are required to follow all local OHRV requirements for PPE and driving safety. Training shall include classroom and in-field instruction as well as a formal driving assessment on an annual basis for each type of vehicle planned for use: i.e. UTV specific training for UTV's and ATV specific training for ATV use. All contractor employees must be fully trained and qualified before use. Proof of individual operator training certifications for each operator shall be available at all times. US DOT rated helmets and safety glasses/goggles are required for any vehicle that does not have a seatbelt and a roll cage. In equipment with a roll cage and seatbelt, operators can utilize a hard hat and chin strap.

4. At the end of each day, unless other arrangements have been made for an extended outage, grounds will be removed and the National Grid project

representative will be notified that all personnel are “clear” of the conductor bus work and equipment.

5. Wherever transmission line workers “touch” wires, a personal ground shall be installed at the work area to establish an equipotential zone, unless workers are engaged in live-line barehand work (29 CFR 1926.964)

7.0 SUBSTATIONS

In addition to the other requirements referenced in this document, this section covers requirements that are specific to substations work.

For additional information, a National Grid representative can provide a copy of the National Grid Substation Maintenance Procedure SMP 499.01.2 *Protective Grounding Procedure* under the Substation Work Methods Infonet page for specifics regarding substation grounding practices.

Grounding plans for substation, major distribution and transmission projects will be submitted to the National Grid construction supervisor a minimum of 1 week prior to construction for review. This plan will show the steps, work area limits and ground cable size and amount. Once reviewed with the National Grid and prior to starting the job, the plan will be reviewed by the contractors with all employees and subcontractors on the project.

The use of an “Equipotential” step/platform or a conductive mat is required for access and egress from the following:

a. Crane or any other equipment, including aerial lift equipment, that is connected to the substation ground grid and/or bonded to transmission line conductors when working outside of the station fence
b. In the rights-of-way
c. In areas inside the substation where there is no ground grid present.

When work is performed inside the substation and there is a ground grid available, the “Equipotential” step/platform or conductive mat is not necessary.

All vehicles shall be grounded and barricaded per OSHA standards and the National Grid Electric Operating Procedure G026 *Mechanized Equipment Grounding*.

Proper clearances shall be maintained from adjacent energized substation bus, energized portions of substation equipment and other transmission lines at all times.

Use of proper insulated tooling (shotguns and sticks) shall be utilized per NECA standard maintaining MAD.

7.1 PPE Requirements

1. All contractors shall comply with the applicable PPE and WZTC requirements referenced in Section 4.0.

2. Contractors who perform any ground breaking activities in a substation within a pre-marked area will require Dig Safe marks to be in place; otherwise, the job must be suspended and the National Grid construction supervisor or project representative shall be notified of the condition.

3. When using non-insulated man-lifts, and if provided by the manufacturer, a secure point of attachment for lifelines, or lanyards or deceleration devices shall be utilized, independent of the means to support or suspend the employee. Workers feet shall also always remain on the floor.

7.2 Notification of Control Authority When Entering a Substation

1. When a contractor enters and exits a National Grid substation, the contractor shall ensure that the System Control Center is notified. While work is being conducted, gates must be monitored at all times or the gates shall be locked. For more information, contact a National Grid representative regarding National Grid EOPG022 *Substation Security*.

2. Unescorted entry in substations can only be provided to contractors who provide assurance that their employees and subcontractors are electrically qualified as specified in 29 CFR 1910.269. Refer to Section 4.0 of this document

3. All National Grid specifically identified bulk power stations will require NERC-CIP training, certification and approval prior to entry to those sites.

7.3 Substation Work Area Identification (SWAI)

1. Contractors who will be working in substations shall follow the SWAI procedure. National Grid will provide a copy of this procedure if required by the project. For more information, contact a National Grid representative regarding National Grid SMP499.10.2 *Substation Work Area Identification Procedure*.

2. Qualified contractors as referenced in section 4.8 of this document shall install their own work area identification. National Grid shall arrange work area identification for non-qualified workers as required.

3. Designated storage areas for items not being used will be posted in the yard and should be the only place these items are kept.

8.0 GAS OPERATIONS WORK

In addition to the other requirements referenced in this document, this section covers requirements that are specific to Gas operations work. For more information, contact a National Grid representative regarding National Grid General Safety Requirements SHE1001 *Gas Policy* which can be found following this link:
http://dc-gasweb1/MelSite/WMSafetyAll.asp .

8.1 PPE Requirements

1. All contractors shall comply with the applicable PPE and WZTC requirements referenced in Section 4.0.

2. The contractor shall wear all appropriate PPE and Class 2 rubber gloves for personal protection when digging or probing within 2 feet of known electrical conductors and when the location of energized conductors is unknown.

8.2 Gas Operations

1. Any contractor who performs covered tasks shall be operator qualified (OQ) as defined in the DOT Title 49 CFR, Subpart N and all applicable state requirements pursuant to the state the contractor is working in. Additionally, any qualifications of contractor employees shall be in full accordance with the Company's Operator Qualification written plan, (OQ Plan) Refer to the most current list of covered tasks in accordance with National Grids' Operator Qualification Program and the Northeast Gas Association, (NGA).

    a. The Operator Qualified status of contractor employees must be regularly updated and accessible through the ISN system. This listing must detail employees' current tasks they are qualified for, the next recertification date, associated documentation and a documented annual acknowledgement in ISN on their qualified workers as referenced in section 3.1 of this document.

b. Contractor personnel involved with covered tasks may require certification by National Grid and an orientation of the involved tasks and National Grid Company standards. National Grid reserves the right to validate contractor qualifications prior to performing Live Gas work.

c. Atmospheres are to be tested with a properly calibrated Combustion Gas Indicator (CGI) or Gas Measurement Instrument (GMI) in accordance with National Grid excavation procedures as required.

d. Each employee in an excavation shall be protected from cave-ins by an adequate protective system, such as sloping, benching or an appropriate shoring system. For more information, contact a National Grid representative regarding National Grid Safety Procedure M-1301 *Standards for Working in Excavations.*

9.0 FORESTRY AND VEGETATION MANAGEMENT

In addition to the other requirements referenced in this document, this section covers requirements that are specific to vegetation management work.

9.1 PPE Requirements

1. All contractors shall comply with the applicable PPE and WZTC requirements referenced in Section 4.0.

2. For work along roads and other areas of vehicular traffic, contractors shall wear high visibility clothing or vests.

3. Flame Resistant Clothing is not required per the applicable OSHA Forestry standard. Forestry contractors must instead wear natural fiber clothing when working within 10 feet of energized equipment.

4. Forestry contractors must wear a properly adjusted full-body fall protection harness connected to an appropriate lanyard when working from an aerial lift. The lanyard must connect to an attachment anchored to either the boom or bucket mounting hardware. Attachment points anchored through only the fiberglass portion of the bucket are not acceptable.

5. Forestry contractors will be required to wear chaps while operating a chainsaw or when assisting and/or working in close proximity to a chainsaw that is being operated.

6. Saws shall not be left unattended with the engine running.

7. When a contractor employee carries a saw, the engine shall be off and/or covered or the saw shall be carried with the blade to the rear and locked.

8. Tree crews will not be allowed to fly their buckets in between the primary and secondary cables if the MAD will be violated in process of doing so.

9.2 Equipment and Work Methods

1. Forestry contractors shall utilize fiberglass sticks and stick saws for work around energized equipment. Additionally, integrity tests shall be performed and documented annually. Test results and expirations shall be available on each vehicle as needed.

2. Forestry contractors shall perform and document dielectric testing of all aerial units annually. Test results and expirations shall be available on each vehicle as needed.

3. For lump sum or unit price mileage trimming projects, a single foreman may supervise up to four (4) bucket trucks on the same project. The minimum qualifications for the "lead" person on each of the other trucks shall be a Journeyman Tree Trimmer or equivalent (Qualified Line Clearance Tree Trimmer). At least one other employee on the truck shall be an OSHA defined, Qualified Line Clearance Tree Trimmer Trainee. For Upstate New York only, it is understood that a Qualified Line Clearance Tree Trimmer shall carry the title, wage and benefits as outlined in IBEW LU 1249's existing contract of a Journeyman Treeman and that a Qualified Line Clearance Tree Trimmer Trainee shall carry, at a minimum, the title, wage and benefits as outlined in IBEW LU 1249's existing contract of a Treeman Trainee, 2nd year.

9.3 Training

1. Forestry contractor management will be required to attend safety council meetings hosted by National Grid as required. The contractor shall ensure that all appropriate safety personnel for the National Grid territory are in attendance.

2. Forestry contractors shall implement and provide the required training and certification programs necessary to provide OSHA defined Qualified Line Clearance Tree Trimmers or Qualified Line Clearance Tree Trimmer Trainees. Qualifications shall be provided in the ISN system. Forestry contractors shall provide an update HASP by April 1st of each year for all work being conducted at National Grid.

3. All contractors using ATV's, UTV's and RTV's for transmission or Forestry work are required to follow all local OHRV requirements for PPE and Driving safety

10.0 LNG PRODUCTION, TRANSPORT AND HANDLING

In addition to the other requirements referenced in this document, this section covers requirements that are specific to LNG Production facilities.

All contractors working at LNG plants will sign in and out of plants daily in the contractors log book. All other gas supply facilities and subcontractors require authorization under the contractor management official. If required by the project, trained National Grid plant personnel shall initially, and as needed, review and re-issue as needed, a work permit process which shall which describe the work being performed, valves with their locations and Lock-out/Tag-out numbers.

10.1 PPE Requirements

1. All contractors shall comply with the applicable PPE and WZTC requirements referenced in Section 4.0 and shall include FR outer clothing.

2. Cryogenic protective gloves/gauntlets and face shields are required when making connections to load / unload LNG. National Grid retains the right to enhance PPE requirements as conditions warrant. The use of additional PPE shall be based on the task performed and the PPE matrix for work in production plants.

10.2 Training

1. Contractors who transport LNG/propane at National Grid facilities are required to be certified in first aid/CPR and are required to complete frost-bite awareness training. Documentation of training records shall be maintained in the ISN system.

2. National Grid expects contractors working at LNG plants to meet the requirements of 49 CFR 193 Subpart H for health, training or experience and/or any applicable National Grid procedures that supersede the above requirements. Contractors shall provide documentation on their qualified workers, as referenced in section 3.1 of this document.

3. All Contractor personnel performing work in LNG plants must meet the requirements of the National Fire Protection Association (NFPA), part 59.

11.0 ELECTRIC GENERATION

In addition to the other requirements referenced in this document, this section covers requirements that are specific to Electric Generation.

11.1 PPE Requirements

1. All contractors shall comply with the applicable PPE and WZTC requirements referenced in Section 4.0

2. Hearing protection is required when working anywhere inside a generation plant and/or outside the plant where noise may be excessive. Acoustic barriers shall be maintained by the contractor as needed.

3. Safety shoes with a minimum height of six-inches are required in Generation plants.

4. Contractors working in generation plants are required to wear 8-Cal clothing protection. For additional guidance, a National Grid representative can provide reference to Electricity Distribution Operations Infonet webpage under Electric Generation's Policies and Procedures EGO-028 *Personal Protective Clothing* & EGO-029 *Personal Protective Equipment*.

11.2 Training

1. Required training may include; PCB's, asbestos, mercury, confined space awareness and excavation competent person requirements. HAZCOM is required by contractors working in generation plants as applicable.

2. Contractors who work at a National Grid Generation Station shall attend an orientation regarding plant safety and as required, US Coast Guard Maritime Security (MARSEC) policies.

3. Equipment training is required per federal, state and local regulations and National Grid procedures. Operators of any powered industrial vehicle must be qualified and documentation shall be documented.

11.3 Equipment & Excavations

1. All excavations shall be performed in accordance with EGO-0005 *Procedure for Excavation in National Grid Generation Facilities* and National Grid Safety Procedure M-1301 *Standards for Working in Excavations.* For additional information, contact a National Grid representative for copies.

2. Gasoline and diesel powered fork trucks shall NOT be used inside the plant or other enclosed facility. Only propane/electric fork trucks are permitted except where additional hazards may exist.

3. All wood products necessary for the work must be made of flame retardant material.

11.4 Equipment Isolation

For isolation of hazardous energy sources while working in Generation plants, please contact a National Grid representative regarding EGO-0010, *Control of Hazardous Energy Sources-Work Permit System.*

12.0 CIVIL CONSTRUCTION

In addition to the other requirements referenced in this document, this section covers requirements that are specific to civil construction work.

12.1 PPE Requirements

1. All contractors shall comply with the applicable PPE and WZTC requirements referenced in Section 4.0.

2. Rubber gloves shall be worn while carrying out work in and around energized or identified direct buried lines, live duct banks, transformer enclosures, manholes, switch gear and other electrical apparatus when performing civil investigations, installations or repairs.

12.2 Enclosed Space Assessment and Ventilation

Contact a National Grid representative regarding the National Grid EOP-UG006 *Underground Inspection and Maintenance* and National Grid Safety Procedure I-902 *Enclosed Space Procedure* for more information regarding enclosed space requirements.

1. Contractors are required to follow all procedures in this document in regards to enclosed spaces (manholes, sidewalk vaults, etc.), including assessment, ventilation, entry and rescue.

2. Each enclosed space shall be tested prior to removing manhole lids and entry. Atmospheric testing shall be continuous for the duration of the entry using a calibrated, industry approved atmospheric tester.

3. When performing hot work or when indicated by atmospheric monitoring, engineering controls such as forced mechanical ventilation shall be used when working in National Grid manholes at all times.

7. All contractors who are qualified electrical workers will treat these spaces as "enclosed spaces" and follow non-entry rescue provisions.

8. In some situations a boom is allowed for retraction from an enclosed space. Refer to Safety Procedure I-902 for more information.

9. Steel cable or wire rope for non-entry rescue is prohibited.

12.3 Equipment Safety Inspection

All contractors shall comply with the applicable equipment safety inspection procedures referenced in Section 5.3

12.4 Excavation Requirements

All excavation work shall be performed under the control of a competent person. All soils in National Grid territories are to be considered class "C", considered unstable and shall require all excavations be performed in accordance with OSHA 1926.651, EGO-0005 *Procedure for Excavation in National Grid Generation Facilities* and National Grid Safety Procedure M-1301, *Standards for Working in Excavations.* For more information, contact a National Grid representative for a copy.

Crews that are performing Excavations shall include an excavation log with their job brief that states the soil type, expected depth and length as well as final depth and length. All required steps need to prevent collapse will be documented on this form as well prior to entry.

Protective systems shall be used for certain manhole installations. These scenarios are covered below:

- The hazard assessment, competent person and/or National Grid supervisor deems it necessary

- If an excavation for a manhole in a roadway is completed and installation of manhole and backfill is not able to be done before the day is complete, a protective system will be required before road plating
- Installation of any manhole 3 way or greater in size/

Where trench boxes are required to be built on site, the contractor shall submit a PE stamped plan and the location shall be designated on the excavation drawings.

All lifts (not limited to materials and equipment) shall be planned and rigged by a competent person. A lift plan shall be provided for all "critical lifts" and must be submitted by a qualified professional to the National Grid representative prior to the lift taking place.

12.5 Cable fault finding and replacements

For excavation work needed to support faulted cables and emergency cable locates, the use of Cable Avoidance Tooling (CAT) shall be used in addition to Dig Safe requirements as an added safeguard to further pin point unidentified buried cables.

For excavations within the tolerance zone, all hand digging in and around direct buried cables shall require basic PPE, non-metallic handled shovels, rubber gloves, FR clothing and EH rated work boots with over shoes.

All excavation equipment shall be grounded in accordance with NG EOP G026. For additional information, contact a National Grid representative.

The use of GPR (Ground Penetrating Radar) shall also be required to verify the Dig Safe/811 locates after award of the project and prior to excavation. This shall include electric URD, UCD and Substation projects.

12.6 Technical Review

Where and when applicable, all trench and excavation work shall be reviewed and stamped by a civil PE in the state of record and will be executed under the supervision of a trenching and excavation competent person. All leading edges of trenches and excavations shall be appropriately demarcated, clearly posted and controlled to prevent unauthorized persons from entering and inadvertently falling into the excavation. All trenches and excavations shall be closed as soon as

practical/possible. All excavations shall be fully controlled for the duration of the exposure by adequately substantial means to withstand the environment and conditions expected to be present.

All pot holing/test holing and exploratory excavations shall utilize vacuum excavation whenever near known underground utilities or hazards, and when the potential for unknown hazards such as live electrical or gas conveyances exist. When using vacuum extraction in combination with air blowing/air knife tools, all persons in the immediate area shall be wearing safety glasses in addition to a full face shields.

13.0 CONSTRUCTION PROJECTS AT CONTAMINATED SITES

In addition to other requirements referenced in this document, all work on contaminated sites must be conducted per the requirements of 29 CFR 1910.120, including the worker qualification and training requirements of 1910.120(e).

14.0 AVIATION

1. Helicopter Crews of two or more shall perform a preflight documented job brief.

2. Helicopter work shall require the use of aviation helmets for both the pilot and passengers.

3. Helicopter pilots and passengers shall participate in the "Flying in the Wire and Obstruction Environment" training prior to flight.

4. Helicopter pilots shall meet the following minimum flight time experience:
    a. 2000 hours as Pilot in Command or Second in command of a rotor craft
    b. 1000 hours in a turbine rotorcraft / helicopter
    c. 100 hours in a helicopter of the make and model to be utilized at National Grid
    d. 300 hours flight time in Wire Environments

For more information, contact a National Grid representative for a copy of EOP T012 *Helicopter Utilization & Notifications.*

15.0 TRANSPORTATION RISKS

Contractor shall define transportation related activities that can have potential process safety consequences. National Grid shall determine if additional risk assessment is needed and contractor shall participate in the assessment.

Contractor shall modify their process to mitigate risk that is determined to be intolerable.

APPENDIX A: NATIONAL GRID CONTRACTOR RISK MATRIX

National Grid Contractor Risk Matrix			
Category	**Description of Work**	**Impact of Work**	**Examples to be included in this category (including, but not limited to)**
Medium / High Risk Exposure **Tier I** **Inclusion in ISN Program is Required**	Physical Work, activity, or service that is performed on National Grid property site or is performed off-site where Owner Client has responsibility and is liable for work performed. Includes, but is not limited to, any activity requiring confined space entry, elevated work, work on operating systems involving hazardous energy, work on contaminated sites, and most work requiring a general work permit, hot work permit, or confined space permit.	Work, activity, or service having: • A potential for causing a catastrophic operational incident; • Access to operations; and/or • A direct role in site operations or maintenance, where failure could result in serious harm to employee or public well-being, company assets, or the environment Also includes any Contractor personnel's job function which has no direct or very limited supervision for operational checks.	• Maintenance, Construction and demolition contractors • Chemical cleaning, tank cleaning • Electricians and Instrumentation Technicians • Movers • Welding • Heavy equipment operations • Well drilling and testing • Environmental investigation, remediation, monitoring activities • Hazardous waste handling and/or transport • Excavation • Food service and handling • Equipment Inspection (e.g., X-ray & other NDT) • On-site sampling / gauging activities (not including escorted storm water sampling) • Common carriers transporting Owner Client-owned LNG or petroleum products • Landscaping services • Snow Removal • Janitorial services

			• Vacuum truck affecting/involving process operations • Oil Spill Response Organizations (OSRO)
Low Risk Exposure **Tier II** **Inclusion in ISN Program is NOT Required**	Work that is office based such as: • Consultants that do not perform work or activities as described in the Medium/High Risk exposure category • Offsite services • On-site vendor pick-up/delivery and repair services • Work performed by public and private utilities • Personnel on-site with Visitor Status, when escorted	Work, activity, or service having an indirect role and no, or limited, access to operations or maintenance where failure could result in serious harm to employee or public well-being, company assets, or the environment.	• Mail/package/part delivery or pick-up (e.g. UPS, Fed EX, vendor-specific) • Samples pick-up by laboratory/courier • Office machine servicing (copiers, printer, computer, etc.) • Laboratory apparatus servicing • Storm water Sampling Labs/Contractors (When Escorted by Owner Client personnel) • Deliver/supply services (vending machine, bottled water, laundry) • Municipal waste pick-up • General trash removal services • Off-site repair/fabrication shops (such as pump, safety valve, piping, vehicle) • Telephone, electric, local municipal utility services • Regulatory representatives • Technical representatives • Engineering services (when escorted by Owner) • Auditors
SVP of SHE can require any contractor to be part of ISN when deemed as a potential risk to National Grid			

EXHIBIT NYSLOF-9

Date of Request: June 14, 2019 Request No NYSLOF-15
Due Date: June 24, 2019 NG Request No. NG-670

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
Case Nos. 19-G-0309 & 19-G-0310
Gas Utilities Rates

Request for Information

FROM: New York State Laborers' Organizing Fund, Harrison Watkins

TO: National Grid, Gas Infrastructure and Operations Panel

SUBJECT: Gas Operations / Maintenance

Request:

Notes: All references to workpapers should be understood to include paper and electronic workpapers, computer-model input or output reports, internal memoranda, studies, or publications. In responding to requests for workpapers, the Company should provide both (1) workpapers created in the process of undertaking calculations; and (2) workpapers relied on to support input or analytical assumptions. When providing electronic spreadsheet files, please leave all cell formulas intact; i.e., do not range value cell formulas of spreadsheets provided. In addition, responses to requests for electronic spreadsheet files should include any linked electronic spreadsheets, as well as all other spreadsheet files or workpapers relied on to develop inputs for the requested spreadsheet file.

For the purposes of these information requests, "Significant Capital Programs" include the Proactive Main Replacement (LPP) Program, the Gas Safety Integrity Management and Integrity Verification: IMP and IVP Programs, the Transmission System Integrity Program, the city/state construction programs (CSC), the Metropolitan Reliability Infrastructure (MRI) Project, and the Renewable Natural Gas (RNG) Interconnections Program.

15. Please describe how the Company validates the competency or training of contractor employees.

Response:

Contractor operator qualifications ("OQs") are maintained by the Northeast Gas Association ("NGA") in its OQ database system ("ITS"). NGA's ITS system tracks contractor qualifications and the Company verifies such qualifications. The Company's Contractor Oversight personnel and Capital Delivery's Contractor Supervisors have access to ITS and validate contractor OQs and monitor ITS for contractor qualification status changes. Both Contractor Oversight and Capital Delivery also perform field verification of contractor OQ.

Capital Delivery Supervisors perform field verification in real time by accessing NGA's ITS system. Contractor Oversight performs real time verification using ITS and may also use a Contractor Oversight Inspection Form.

Daily job briefs may also include field verification of qualifications related to specific tasks.

Additionally, planned improvements to the Company's contractor OQ verification and onboarding processes are described in Attachment 1. Slide 4 describes the Company's new process for verifying OQ through ITS using "e-wallet" cards.

Name of Respondent: Frank Prost

Date of Reply: June 24, 2019

Operator Qualification Program Enhancement Plan

May 14, 2019



nationalgrid

Agenda - OQ White Paper Best Practices Status Update

- Written exam security enhancement
- Employee training & OQ verification process
- OQ program enhancement
- Performance evaluation enhancement
- Training & evaluation process for personnel performing Engineering tasks
- Contractor training & qualification enhancement
- On-boarding contractor process
- Effectiveness of OQ program
- Gas technical training transformation project deliverables

Written Examination Security Enhancement

Significant investment has been made to achieve the written exam security enhancements, which are now the most comprehensive in the industry:

- Execution of $1.3M contract with Prometrics for exam security – February 2018
- Prometrics testing sites in DNY training centers
  - Hub Drive, Melville - In operation since 12/2018
  - Springfield, Queen – In operation since 01/2019
- Prometric testing sites in UNY training centers
  - Seneca Street, Schenectady - In operation since 12/2018
  - Henry Clay Blvd., Liverpool - In operation since 01/2019
- Additional Prometric sites are under construction at:
  - DNY – Gulf Avenue, Staten Island – 15 May, 2019
  - UNY – Henry Clay Blvd. Ops#1, Liverpool – 12/2019
- Rewrote all written exams and implemented all new exams - 2017

Employee Training & OQ Verification Process Enhancement

- **Execution of $1 M contract with Industrial Technical Service (ITS) – March 2018**
- **OQ written exams are performed in ITS – OnBoard system at our Prometrics secure test sites**
- **All employees carry an eWallet Card**
- **Each eWallet card has a employee picture and unique QR code that allows employees, supervisors, and regulators the ability to access the card holder's Operator Qualifications on demand using any QR reader app from smart phone**
- **The OnBoard system, where National Grid Operator Qualifications are managed, has the functionality to automatically align Skills to Tasks to quickly identify compliance.**





OQ Program Enhancement

- **Changed all 5-year requalification intervals to 3-year intervals**
- **Addition of seven new performance evaluations including exothermic welding, leak survey, line locating, regulator station inspection and three compressor station related tasks (total of 23 covered tasks today)**
- **Reviewed existing covered tasks and subtask with written exam only. Based on complexity and associated risk, we are considering additional 30 covered tasks that would require performance evaluation. Out of 72 covered tasks, 53 covered tasks and subtasks would require performance evaluation along with written exam.**
- **Availability of experienced evaluators for performance evaluation is a concern**
- **National Grid has adopted more conservative field Span of Control than ASME B31Q (1:2 for non-critical function)**
- **Inclusion of Task-specific abnormal operating conditions (AOCs) in the domain of content covered by written evaluations**
- **Subject Matter Expert (SME) review and update of Covered Task domains, elements, and AOCs**
- **Update to the OQ management of change process**

Performance Evaluation Enhancement

- **All performance evaluation observation steps will include hold points**
- **Each observation steps and final pass/fail results records will reside in ITS OnBoard system**
- **Each performance evaluation requires evaluator signature, student's signature, date of evaluation and reasons for failure**
- **Performance evaluation observation steps are currently reviewed by SME to ensure that each performance evaluation contains AOC related question to check the knowledge and skill of student to recognize and react to AOC**
- **Incorrect answer to AOC related question during performance evaluation is considered failure of examination**
- **100% passing score required for performance evaluation**
- **All performance evaluations are performed one-on-one except for plastic joining qualification. One evaluator observes four students performing manual butt fusion, electrofusion, heat saddle fusion and mechanical joints in line of sight inside the laboratory**
- **Butt fusion joining qualification using hydraulic machine evaluation is performed on one-on-one basis**

Training & Evaluation Process for Personnel Performing Engineering Tasks

- **Model plan requires process for personnel performing engineering tasks shall be added to the OQ program so that functions such as the development of design and engineering modifications to the pipeline system are performed and reviewed by qualified person.**
- **Design drawings and procedures will be reviewed with the stakeholders prior to construction and signed by engineer**
- **Additional peer review process for design and engineering projects will be incorporated into standard practices**

Contractors Training and Qualification Enhancement

- **To ensure contractors employees are familiar with the company specific O&M procedures, tools and equipment, they are required to pass the National Grid challenge test.**
- **Currently challenge tests are administered in DNY only. We are rolling out the challenge program in UNY in 2020.**
- **Any new trained and qualified contractor employee allowed to work on National Grid system only if successfully demonstrate the skill through challenge to perform specific function that may include several covered tasks.**
- **Challenge tests are required for following tasks:**
  - PE joining including heat, electro fusion and mechanical joining
  - Drilling, Tapping & Stopping off using company approved tools, equipment and procedure on PE, Steel and CI distribution system
  - Service installation including meter and regulator header build up
  - Tie in and flow control
  - Steel welding using company approved Welding Procedure Specification
- **Contracted employees will be required to take a supplemental written test that would cover exam questions related to National Grid O&M procedures and equipment**

On-boarding Contractors

- **Our current protocol is to review, assess and evaluate the contractors training and qualification written program prior to awarding the contract to a qualified contractor**
- **Our evaluation process includes reviewing credentials of contractor trainers, their actual operator qualification, experience and knowledge of our company specific O&M procedures, Work Methods, Standards, Gas System and Tools & Equipment**
- **Our contract oversight personnel frequently visit contractors training centers, classrooms and mock-up shops to witness, observe and verify training being offered to their employees**
- **We also perform audits of contractors training and qualification programs**
- **We are working jointly with NGA, LDC's and contractors to develop a recommended practice (RP) procedure to ensure that contractor training & qualification programs meet company specific requirements**

Effectiveness of OQ Program

- **To ensure that Qualified personnel are performing covered tasks in compliance with our company procedures, our quality inspectors, company inspectors and compliance analysts perform unannounced random inspections at job sites and perform audits of records of plastic fusion inspections, pressure testing, operator qualifications and as-built maps.**
- **We have a selective re-dig program in place for recently completed field work for many years. This program is administered by our quality assurance organization to check the conformance to our company standards and procedures for critical high-risk tasks.**
- **Any deficiency found during inspection requires a comprehensive incident analysis (IA) process to determine the root cause and corrective actions. If an employee workmanship is the root cause of an incident than an employee OQ is revoked and requires retraining and retesting.**

Gas Technical Training Transformation Project

- **3+ year program to revise all gas technical training for field workforce**
  - Define employee competency by role
  - Revise curriculum content and training methods
    - Defined classroom, hands-on practice, and OJT for each Job Family role
    - Fully aligned with policies and procedures
  - Build instructor capability from 'hire to retire' and improve overall training academy efficiency
  - Incorporate new technology into training operations
  - Link NG revised curriculum to contractor training and qualification
- **Job families prioritized to focus on largest workforce areas first**
  - Construction & Maintenance, Customer Meter Services underway now, first curriculum revisions expected in late 2019
  - Instrumentation & Regulation, Corrosion, and Damage Prevention job families to follow on in 2020
  - Revised Curriculum to be shared with Contractors to ensure equivalent training materials used

EXHIBIT NYSLOF-10

CONSOLIDATED EDISON COMPANY OF NEW YORK, INC.

STANDARD TERMS AND CONDITIONS

FOR

CONSTRUCTION CONTRACTS

October 15, 2014

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Appendix A - Required Clauses and Certifications

STANDARD TERMS AND CONDITIONS FOR CONSTRUCTION CONTRACTS

1. Definitions. The following terms as used herein shall have the meanings stated:

"Con Edison" - Consolidated Edison Company of New York, Inc., the entity entering into the Contract and issuing any purchase orders applicable to the Contract, for work to be performed for Con Edison or its affiliate Orange and Rockland Utilities, Inc. ("O&R").

"Contractor" - The contractor who is a party to the Contract with Con Edison .

"Contract" - The contract between Con Edison and Contractor consisting of: (a) a Blanket Purchase Agreement ("BPA") or Contract Purchase Agreement ("CPA") and/or a Con Edison Standard Purchase Order ("purchase order"); (b) the relevant Con Edison request for quotation; (c) these Standard Terms and Conditions; and (d) any documents or portions thereof incorporated by reference in (a), (b) or (c) above, including, but not limited to, special conditions, specifications, performance requirements, plans and drawings. The words "hereof," "herein," "hereto" and "hereunder" as used in these Standard Terms and Conditions shall be deemed to refer to the Contract.

"Work" - The project contemplated by the Contract and all labor and supervision; construction materials, equipment, tools and other aids to construction; equipment, materials and structures to be installed; and other things of any nature necessary or proper for the completion of the project, whether or not expressly specified herein.

"Subcontractor" - Any company or person, other than an employee of Contractor, that furnishes any of the Work on behalf of Contractor.

2. Contract Formation. A legally enforceable agreement shall arise upon the signing or acknowledgement electronically by Contractor of the Contract in Oracle E-Business Suite iSupplier (the "Procurement System") or, if Contractor is not enabled in the Procurement System, upon the mailing or delivery by other means of the Contract or another writing manifesting acceptance of Contractor's offer; provided, however, that if Contractor's offer contains terms additional to or different from the terms on which quotations were requested by Con Edison which are not accepted in writing by Con Edison, a legally enforceable agreement shall not arise until the signing or acknowledgement electronically by Contractor of the Contract in the Procurement System or, if Contractor is not enabled in the Procurement System, by the signing by Contractor of the Contract or a copy of the Contract or such other writing as may be issued by Con Edison (or another document expressing Contractor's acceptance thereof), or Contractor's commencement or continuation of the Work following its receipt of the Contract or such other writing, such Work signifying Contractor's acceptance of the terms thereof.

3. Specifications, Plans, and Drawings.

A. The Work shall be performed in strict accordance with the Contract specifications, plans, and drawings. No deviation is permitted unless approved in advance in writing by Con Edison. The specifications, plans, and drawings and all other documents which are part of the Contract are supplementary to each other. Anything called for by any one of such documents shall be required to the same extent as if called for by all of them, and the Work shall be completed in every detail whether or not every item is specifically mentioned. If there should be a conflict between the drawings and the writings that comprise the plans and specifications, the writings shall govern unless upon notice thereof Con Edison directs otherwise in writing.

Contractor shall carefully review all of the contract documents. Contractor shall promptly submit in writing to Con Edison whenever discovered, whether before or after award, any inconsistency, ambiguity, or error between documents or within a document. Contractor shall abide by the written direction of Con Edison's authorized representative with respect to such matters, which direction shall be final and binding.

Contractor waives any claim for extra compensation based on an inconsistency, ambiguity, or error which Contractor could have discovered by reasonable diligence and prudence.

B. Con Edison may furnish to Contractor any additional plans, drawings, specifications or other documents which it considers necessary to illustrate or explain the Work in further detail, and Contractor shall comply with the requirements of all such documents.

C. Contractor shall, throughout the time during which the Work is being performed, keep at the work site available for inspection by Con Edison one complete and current set of the Contract documents, including, but not limited to, the Contract plans, specifications, and drawings, any additional documents furnished by Con Edison, and all shop and work drawings approved by Con Edison.

D. Where required by the Contract, Contractor shall submit designated documents, such as drawings and process procedures, for review and approval by or on behalf of Con Edison. All proposed changes to or deviations from such documents after they have been approved shall also be submitted to Con Edison for such review and approval prior to their implementation.

4. Price and Payment.

A. Unless expressly stated to the contrary herein, all prices are firm and not subject to increase. All payment periods, including discount periods, shall

begin upon receipt of proper invoices by Con Edison's Accounts Payable Department. Unless otherwise specified in the Contract, payment shall be made within thirty (30) days of receipt of each invoice. Payments by mail shall be deemed made when deposited in the mail. Notwithstanding anything to the contrary herein, no invoice submitted for a partial or progress payment shall be processed unless and until Contractor furnishes to Con Edison a Contractor Affidavit – Partial Release And Waiver of Lien and for each subcontractor, a Subcontractor Affidavit – Partial Release of Lien, duly executed and delivered by Contractor and its Subcontractors at issue, as applicable. Such documents shall, among other things, state that each statement set forth in paragraph (H) of this Article is true and correct concerning the invoice and the amount requested therein, and, with respect to Work covered by or included within the invoice, waive, release and discharge claims and liens, contain a covenant to pay and release of record, or otherwise discharge of record, all liens and contain a covenant to defend, indemnify and hold harmless Con Edison and its affiliates (including, but not limited to, O&R), and any owner of the real property on which the project is situated), from and against such claims and liens and any related costs and expenses. Such documents shall be in a form reasonably satisfactory to Con Edison and shall in all respects be read and interpreted consistent with Section 34 of the New York Lien Law (or its successor).

No invoice submitted for a final payment shall be processed unless and until: (i) Contractor has fully performed all the Work to Con Edison's satisfaction; (ii) Contractor has delivered to Con Edison all warranties, manuals, operating instructions, drawings, and all other documents required by the Contract; and (iii) Contractor has submitted documents sufficient to satisfy Con Edison that all the Work has been properly performed, that payment is due to Contractor, and that all Subcontractors who performed or furnished labor, materials, supplies, or equipment for the Work have been fully paid, or that they will be paid promptly from monies received from the final payment. The documents required together with the invoice for final payment shall include a Contractor Affidavit – Final Full Release And Waiver of Lien document and, for each Subcontractor, a Subcontractor Affidavit – Final Full Release And Waiver of Lien document, duly executed and delivered by Contractor and its Subcontractors, as applicable. Such documents shall, among other things, state that each statement set forth in paragraph (H) of this Article is true and correct concerning the invoice and the amount requested therein and, with respect to the Work, waive, release, and discharge all claims and liens to the extent permitted by law, contain a covenant to pay and release of record, or otherwise discharge of record, all liens, and contain a covenant to defend, indemnify, and hold harmless Con Edison and its affiliates (including, but not limited to, O&R) and any owner of the real property on which the project is situated from and against all such claims and liens and related costs and expenses. Such documents shall be in a form reasonably satisfactory to Con Edison and shall in all respects be

read and interpreted to be consistent with Section 34 of the New York Lien Law (or its successor).

B. Unless otherwise provided in the Contract, for unit price Work involving layouts, Contractor shall submit invoices upon completion of layouts or layout parts. For other unit price Work, invoices shall be submitted monthly. For all unit price Work, the judgment of Con Edison as to the quantity of Work completed and whether or not it is acceptable shall be conclusive.

C. For lump sum Work, Contractor shall render progress payment invoices monthly unless stated otherwise in the purchase order. Each invoice shall be based on an estimate, certified by Contractor and approved by Con Edison, of the physical Work performed during the period stated in the purchase order or, if none is stated, during the preceding calendar month. In preparing estimates, Contractor shall include up to 75% of the cost of conforming materials delivered onto the site but not yet physically incorporated into the Work. As additional security for the proper performance of the Contract, Con Edison will retain ten percent (10%) of the amount of each invoice until fifty percent (50%) of the Work is completed. Thereafter, if the Work is progressing satisfactorily and on schedule and Contractor is in compliance with all of its obligations hereunder, Con Edison shall pay the remaining invoices for progress payments in full. The amount previously retained will be paid to Contractor following completion and acceptance of the entire Work. The judgment of Con Edison as to the value of the Work completed, whether it is on schedule and whether it is acceptable shall be conclusive.

D. Con Edison shall have the right at any time to withhold from any payment which may be or become due under the Contract such amount as may reasonably appear necessary to it to compensate Con Edison for any actual or prospective loss due to Work which is defective or does not conform to Contract requirements, actual or prospective failure of Contractor to complete performance of the Work, or any other failure of Contractor to perform any of its obligations under the Contract or when it reasonably appears to Con Edison that Contractor has previously been overpaid. Con Edison shall be entitled to retain any and all amounts so withheld until Contractor has, in Con Edison's judgment, either performed the obligation or obligations in question or furnished security which Con Edison deems adequate for such performance or, in the case of withholding for overpayment, until an audit of Contractor's work or documentation is completed and the proper payment is determined.

E. Con Edison at any time may, after notifying Contractor in writing, pay directly any unpaid claims against Contractor based on the Work, and in so doing Con Edison shall be conclusively deemed to be acting as Contractor's agent. Any payment made by Con Edison to discharge a claim against Contractor shall be treated as a payment made under the Contract from Con Edison to Contractor.

F. Con Edison shall not be liable to Contractor for interest on any late payments unless expressly provided for herein. If for any reason Con Edison is in arrears in payment(s) hereunder, payment(s) made shall be applied by Contractor to any and all principal sum(s) due before being applied to any interest that may be due thereon pursuant to any express provision therefor in the Contract or otherwise.

G. Except to the extent specified in a written reservation of rights, the acceptance by Contractor of final payment shall be and shall operate as a release of all claims against Con Edison and of all liability of Con Edison to Contractor for things done or furnished in connection with the Work and for every act and neglect of Con Edison and others for whom Con Edison may be responsible relating to or arising out of the Work. However, no payment, final or otherwise, shall operate to release Contractor or its sureties from the obligations under this Contract or any performance or payment bond.

H. Contractor shall submit invoices in a form reasonably acceptable to Con Edison. Each invoice shall constitute a representation by Contractor that (i) the payment requested therein reflects Work performed and costs incurred by Contractor on account of such Work, (ii) the materials, supplies and equipment for which such invoice is being submitted have been installed or incorporated in the Work or have been stored at the project site or at such off-site storage locations as shall have been approved in writing by Con Edison; provided, however, that Con Edison shall not be obligated to pay or advance monies for materials stored off-site, unless approved in advance and in writing by Con Edison, (iii) the materials, supplies and equipment, if stored off-site, have been properly insured and stored in a "bonded" warehouse and are insured in accordance with this Contract and, if stored at the project site, are insured in accordance with the provisions of this Contract, (iv) the materials, supplies and equipment are not subject to any liens or encumbrances, (v) no mechanic's, laborer's, vendor's, materialman's or other liens or payment bond claims have been filed in connection with the Work for any of the labor, materials, supplies or equipment incorporated therein or purchased in connection therewith, (vi) all Subcontractors who performed or furnished labor, materials, supplies, or equipment for the Work for which payment is requested or for prior Work have been fully paid, or that they will be paid promptly from monies received from such invoice, (vii) the Work which is the subject of such invoice has been performed in strict accordance with the Contract documents and all applicable legal requirements, and (viii) the amount of the partial payment for which the invoice is submitted, when added to the sum of all previous payments made under prior invoices, does not exceed that portion of the price under the Contract which is allocable to the portion of the Work that has actually been completed on or before the date of such invoice and that the remainder of the price under the Contract (as the same may have been adjusted hereunder) is sufficient to pay in full the costs necessary to

perform and complete the Work. Upon Con Edison's request and as a condition precedent to payment, Contractor shall provide supporting documentation that is satisfactory to Con Edison to substantiate the representations set forth in this paragraph.

5. Time for Completion. TIME IS OF THE ESSENCE OF THE CONTRACT and of each and every portion thereof for which a certain length of time is fixed for performance. Unless otherwise specified, Contractor shall start the Work at the construction site within ten (10) days after it receives a written Notice to Proceed from Con Edison and shall complete the Work and the various parts thereof within the time or times specified in the Contract schedule. Contractor shall prosecute the Work regularly, diligently and without interruption at such rate of progress as will ensure completion within the specified time, and shall furnish properly skilled workmen and materials, tools, and equipment in numbers and amounts sufficient to accomplish this. Contractor agrees that the time or times specified for completion of the Work and of any part thereof are reasonable, taking into consideration all facts and circumstances. Work specified by the Contract to be performed after regular hours or on Saturdays, Sundays or legal holidays and Work performed at such times as a result of conditions in a permit or because of local regulations or to adhere to or regain the rate of progress required by Con Edison shall be performed without additional expense to Con Edison. If, in the opinion of Con Edison, Contractor falls behind schedule, Con Edison shall have the right to require contractor, at no additional cost to Con Edison, to increase its labor force or days or hours of work, to work overtime or increase the number of shifts, to use additional equipment or other construction aids, or to take such other steps as may be necessary to ensure completion of the Work on schedule. Receipt and acceptance by Con Edison of revised schedules from Contractor during the Work shall not be deemed a waiver of the schedule initially approved.

6. Excusable Delay. Contractor shall be excused any delay in completion of the Work arising from a cause beyond its control which it could not with the exercise of due diligence have either foreseen or avoided, including act of governmental authority, act of God, extraordinary weather conditions, flood, accident such as fire or explosion not due to the negligence of Contractor, strike which is not the result of an unfair labor practice or other unlawful activity by Contractor, riot, failure of public transportation facilities, inability of Con Edison to provide access due to plant malfunctions, and inability to perform caused solely by Con Edison's act or failure to act in breach of an express obligation under the Contract. Delay in Contractor's receipt of subcontracted supplies or services for reasons beyond the control of the Subcontractor shall not be excusable delay hereunder to the extent that the supplies or services are available to Contractor from another source. The unavailability of sufficient, qualified labor to perform the Work shall not be excusable delay hereunder unless the unavailability is caused by a strike which is not the result of an unfair labor practice or other unlawful activity by Contractor. Contractor shall give written notice and full particulars of the cause of any delay within 48 hours after its occurrence and thereafter shall update Con Edison on a bi-weekly basis. The time for performance in any such instance shall be extended by a period equal to the time lost by reason of the excusable delay. Such extension shall be Contractor's sole and exclusive remedy for such delay and Con Edison shall not be liable for any damages or additional costs incurred as a result of such delay.

7. Safeguards in Work.

A. Contractor shall provide and maintain at its own expense safe and sufficient entrance and exit ways, walkways, platforms, barricades, warning lights, scaffolds, ladders, runways for concrete carriers, hoists and all equipment, apparatus and appliances necessary or proper for carrying on the Work safely; shall not load any of the foregoing items or any part of any structure or equipment with a weight that will make it unsafe; shall make and keep the place of Work and the ways and approaches thereto well lighted, safe and free from avoidable danger, taking into account, without limitation, local conditions; and shall mark any faulty items "unsafe" until repaired or replaced.

B. Contractor shall provide all permanent and temporary shoring, anchoring and bracing required by the nature of the Work to make all parts absolutely stable and rigid, even when such shoring, anchoring and bracing are not explicitly called for. Contractor shall support and protect all buildings, bridges, roadways, conduits, wires, water pipes, gas pipes, sewers, pavements, curbing, sidewalks, fixtures and other public or private property that may be encountered or endangered in the prosecution of the Work.

C. In accordance with the rest of this Article 7 and without limitation thereof, Contractor shall test all areas, excavations, openings, manholes, vaults and boxes, for an adequate supply of oxygen and for any and all toxic, harmful or combustible gases or fumes or other dangerous substances before and during the course of the Work and shall provide all the necessary equipment, including, but not limited to, all oxygen deficiency and gas testing apparatus required for such tests.

D. Contractor shall strictly observe safety requirements of applicable federal, state and municipal laws and regulations, including, without limitation, the Federal Occupational Safety and Health Act. Contractor shall cause all equipment and structures, the place of Work and the ways and approaches thereto to meet the requirements of all public authorities. Contractor shall comply with the requirements of and recommendations in the latest edition of the "Manual of Accident Prevention in Construction," published by The Associated General Contractors of America, to the extent that such provisions are not inconsistent with other provisions of the Contract or applicable laws or regulations. Contractor shall maintain an accurate record of all cases of death, occupational disease or injury requiring medical attention or causing loss of time from work arising in connection with performance of the Work.

E. If in the opinion of Con Edison's authorized representative, Contractor's work practices or conditions created by Contractor are unsafe or fail to comply with applicable laws or regulations, Con Edison may halt the work until such practices and conditions are corrected. Contractor shall not be

entitled to any additional costs or time for performance due to such work stoppage. If, when the Con Edison authorized representative is not present at the site, a Con Edison employee (or O&R employee for Work ordered for O&R) directs Contractor to discontinue an operation because it may be unsafe or illegal, Contractor shall immediately halt the questioned operation and, if the Contractor disagrees with the employee, shall contact the Con Edison authorized representative for instructions. Contractor shall obtain the employee's name and employee identification number and report this information to the Con Edison authorized representative.

F. Contractor shall be responsible for any failure or neglect on its or its Subcontractor's part to perform the obligations contained in this article, and shall defend and indemnify Con Edison and its affiliates (including, but not limited to, O&R) against any loss, liability, damage or expense resulting in whole or in part from such failure or neglect.

G. If the Work involves pipeline facilities for the transportation of gas, hazardous liquids or carbon dioxide or a liquefied natural gas facility subject to Part 192, 193 or 195 of Title 49 of the Code of Federal Regulations (CFR), Contractor shall comply, and shall require its employees to comply, with the drug and alcohol testing requirements of 49 CFR Part 199. Contractor shall maintain and follow written anti-drug and alcohol misuse plans and shall provide the testing, education, and training required by the Regulations. Contractor shall allow access to its property and records concerning the plans and their implementation to Con Edison, O&R and their representatives, to the Department of Transportation Administrator, and to representatives of federal or state authorities having jurisdiction for the purposes of monitoring compliance with these requirements.

8. Knowledge of Work Conditions and Requirements. Contractor represents that it has visited and examined the site of the Work and satisfied itself as to the general and local conditions, particularly those relating to transportation, handling and storage of materials, availability of labor, water, drainage, power, roads, weather, ground and other physical conditions at the site, and as to all other matters which could affect the Work or the cost thereof. Contractor also acknowledges that it has examined the specifications, drawings, and other Contract documents and has satisfied itself as to the requirements of the Work, and has seen or had an opportunity to ask about all conditions which may affect the Work, including equipment or structures in place or to be in place, or work being or to be performed, which could interfere with the uninterrupted performance of the Work. Contractor agrees that its entry into the Contract has not been induced either wholly or in part by any promises, representations or statements on behalf of Con Edison other than those set forth in the Contract, and that any failure of Contractor to examine the Work site, Contract documents or all other available information shall be at its own risk. Contractor further represents that the price set forth in the Contract has been determined with due regard by Contractor to all such conditions and requirements affecting the Work, as well as the difficulties and delays incident to work of the nature contemplated hereby, and agrees that no claim for any increase in such price shall be made except as specifically provided in the Contract.

9. Contractor's Performance.

A. Contractor shall perform in good workmanlike manner and in accordance with best accepted practices in the industry all the Work specified or reasonably implied in the Contract, in accordance with its terms and the directions of Con Edison and its authorized representatives as any may be given from time to time. Contractor's performance shall include, except as otherwise specifically stated in the Contract, everything requisite and necessary to complete the Work properly, notwithstanding the fact that not every item involved is specifically mentioned, including, but not limited to all materials, labor, tools, equipment, apparatus, water, lighting, heating, power, transportation, superintendence, temporary construction, site security and all other services and facilities of every nature necessary or appropriate for the execution of the Work on schedule. Details which are not specified in the Contract shall be performed by Contractor at no extra cost if such details are within the general description of the Work.

B. When work is performed outside of Contractor's own premises, Contractor must provide at all times an on-site representative with full authority to act for Contractor. The representative must be able to read, write, and thoroughly understand both English and any other languages spoken by persons performing work for Contractor and must be able to effectively communicate with those persons in their own language or languages. In addition, Contractor's representative must ensure that labeling, log book entries, completion of forms and all other tasks requiring a proficiency in English are performed clearly and correctly. The continuation of the individual selected by Contractor in this role shall be subject to the continuing approval of Con Edison.

C. Contractor shall perform the Work in accordance with the following:

(i) All equipment, tools, other construction aids and materials utilized by Contractor shall be of high quality and in good working order. Contractor shall submit material safety data sheets (MSDS) for all chemical and hazardous substances used in the Work. If, in the opinion of Con Edison, any of Contractor's equipment, supplies, tools, other construction aids or materials are unsafe or inadequate, Contractor shall remove such items from the site immediately and replace them with safe and adequate substitutes at Contractor's expense. Contractor shall be fully and solely responsible for and shall safeguard its equipment, tools, supplies, other construction aids and materials at all times. Contractor shall provide adequate storage for all such items used in connection with the Work.

(ii) The use of public roadways and properties for the parking of employee vehicles, construction equipment, receiving and placement shall be in accordance with the applicable laws and ordinances. Access to all underground facilities, as for example through Con

Edison, municipal and telephone company manholes, shall be maintained and allowed during the entire performance of the Work. Adjacent private properties shall not be entered or used for any such purpose without the written consent of the property owners.

(iii) Fire hydrants and stop valves adjacent to the Work shall be kept clear and readily accessible to fire apparatus, and no material or other obstruction shall be placed, parked or stored within fifteen (15) feet of any hydrant or stop valve (or a greater distance if required by local law, rule or regulation). Contractor shall comply fully with all local rules and regulations relative to fire protection, shall keep the structure and premises free from burnable trash and debris, and shall exercise every precaution against fire. This shall include, but not be limited to, posting a fire watch, with appropriate fire fighting equipment, during all welding, burning, stress relief and other heating operations. Contractor shall assure that the fire watch is informed of the site fire control procedures and remains posted during breakfast, lunch and dinner periods and until one hour after such heating operations have stopped.

10. Con Edison Authority

A. Con Edison shall have the authority to decide any and all questions which arise in connection with the Work, and Con Edison's decisions shall be conclusive and final. Con Edison shall be the final judge of the meaning and intent of the Contract and all provisions thereof. Con Edison shall have the authority to conclusively resolve any disagreements which may arise between Contractor and any other contractor.

B. Con Edison shall have the right to maintain a representative at the construction site. Such representative may, on request, give Contractor reasonable assistance in interpreting the Contract drawings, specifications and plans, but such assistance shall not relieve Contractor of any duties (including, without limitation, those of giving notice to or securing the approval of Con Edison) arising under the Contract.

11. Estimated Quantities. Whenever estimated quantities of Work to be done on a unit-price basis are shown in the Contract, differences between the actual number of units of Work encountered and the estimated quantities of units shall not result in an increase or decrease in the unit prices or provide the basis for any claim against Con Edison or O&R by Contractor.

12. Warranties.

A. Contractor warrants the Work for a period of three (3) years from the date of completion and acceptance of all Work, unless a longer period is specified elsewhere in the Contract or in an applicable municipal code or regulation, in which case the longer period shall apply:

(i) as to services, that they shall be rendered competently and by qualified personnel in accordance with the best accepted practices in the industry and comply strictly with all specifications and performance requirements contained in the Contract; and

(ii) as to materials, equipment, structures and other things, that they shall be new and free from defects in title, material, workmanship and design, conform strictly to all applicable specifications, and be suitable for their intended use. Contractor further warrants that the Work shall meet any and all tests and comply strictly with all specifications and performance requirements contained in the Contract. The warranty of good title shall be unlimited in time.

B. In the event any part or all of the Work fails to satisfy any of these warranties, upon written notice thereof from Con Edison, Contractor shall, at no cost to Con Edison, promptly repair, replace, or reperform the defective Work, as directed by Con Edison, and do whatever else is necessary to cause the Work to satisfy all of the aforesaid warranties. All work repaired, replaced or reperformed under the provisions of this Article shall be subject anew to this Article with the warranty period commencing upon completion of the repair, replacement, or reperformance. If Contractor fails to correct any defective Work as aforesaid promptly after being notified thereof by Con Edison, then Con Edison may, at its option, either correct the defective work and charge Contractor for the costs and expenses it incurs in so doing or secure an equitable reduction in the Contract price based on its retention of the defective Work. Any defective parts removed in connection with repair or replacement shall be disposed of by Contractor at its expense.

C. In addition to making the foregoing warranties, Contractor agrees to obtain, for the benefit of Con Edison, from any and all Subcontractors hereunder, the same warranties as those required of Contractor under this Article and not to accept any warranties which are inferior in any respect to those required under this Article.

D. All warranties made or obtained hereunder are made to, and for the benefit of, Con Edison and O&R and may be enforced by or on behalf of either or both of Con Edison and O&R.

13. Changes (Including Extra Work).

A. Con Edison shall have the right at any time, by written notice (electronically or in print form) to Contractor and without notice to any of Contractor's sureties, to direct changes in the Work, including direction to do extra work or work outside of normal hours (when such work is not already the responsibility of Contractor under the Contract) or to delete part of the Work. If any such change causes an increase or decrease in the cost or time required for performance hereunder, an equitable adjustment shall be made

in the Contract price or schedule, or both, as follows: If the change involves work of the kind for which unit prices are contained in the Contract, such work shall be paid for at those prices unless Con Edison at its discretion determines not to use such unit prices and so notifies Contractor prior to the start of the changed work. For work for which no unit price is established or for which Con Edison has determined not to apply the unit prices, the parties shall endeavor to agree on a lump sum price for the change. If the change is not defined well enough for a fixed price, or if there is not enough time to negotiate one, or if the parties do not agree on one, Contractor shall perform the change on a time-and-materials ("T&M") basis at rates for labor, equipment and materials approved by Con Edison. In the case of deletion of any portion of the work for which no unit price is established, the Contract price shall be reduced by the decrease in Contractor's cost of performance and profit thereon.

B. The following rates are approved by Con Edison for Work performed on a T&M basis:

(i) Labor

Contractor will be compensated for each hour performed at the straight and premium time rates (as applicable) specified in the applicable schedule of rates. With respect to Work ordered for Con Edison, the applicable schedule of rates will be the Con Edison schedule of rates entitled "New York City and Westchester County Labor Rates for Time and Materials Work" or "Maintenance Agreement Labor Rates -- New York City and Westchester County," as applicable, in effect at the time the Work is performed, except that an additional 10% will be paid for the straight time portion of labor performed by one approved Subcontractor. With respect to Work ordered for O&R, the applicable rates will be the rates agreed to in the Contract. The labor of superintendents, non-working foremen, timekeepers and clerical employees is not compensable. This paragraph shall not be construed to affect Contractor's obligations to pay its employees in accordance with applicable law and the requirements of the Contract.

(ii) Materials

Contractor will be compensated for all materials used for the Work at the actual net cost plus 10%. Contractor shall furnish Con Edison with the invoices evidencing the purchase of such materials (and any other back-up documentation).

(iii) Equipment

Contractor will be compensated for equipment employed in the Work at 70% of the rates, including operating costs, in effect at the time the Work is performed, in the "Rental Rate Blue Book for Construction Equipment" or

the "Rental Rate Blue Book for Older Construction Equipment," whichever is applicable.

All rates are based on 8 hours per day, 40 hours per week, 176 hours per month and 22 working days per 30 day period. The rate resulting in the lowest cost to Con Edison for the employment of the equipment in the Work will be used. Operating costs will only be paid for hours during which equipment is actually being used in the performance of the Work.

Each unit of equipment or tool with a value at the time of use of less than $500 will be considered included in the labor rate and will not be separately compensable.

C. Contractor shall not perform any changes in the Work, including extra work, except pursuant to written direction from Con Edison's representative authorized to make changes expressly and unmistakably indicating his intention to change the Work. In the event adjustments for such changes are not agreed upon promptly, Contractor shall nevertheless proceed diligently to effect the changes at the time it is directed to do so by Con Edison.

D. In the absence of a written direction described in the preceding paragraph C, if Contractor deems any directive, whether oral or written, by Con Edison's authorized representative to be a change in the Work, Contractor shall nevertheless comply therewith but shall within five days give written notice as described in Article 16, Claims, and comply with the requirements thereof.

E. Any price increase or decrease or extension or acceleration of time for performance shall not be binding on Con Edison unless evidenced by a Contract modification or change order signed and issued by Con Edison. Contractor shall not have the right to make changes in the Work without the prior written approval of Con Edison.

F. Prices agreed upon for, or applicable to, changes (including extra work), include all impacts of the changes on the Work, including, but not limited to, delay, loss of productivity, demobilization, remobilization and idle time, and Contractor shall have no other claim for other effects on the Work due to such changes.

14. Labor.

A. Contractor shall, unless otherwise specifically stated herein, provide all labor required to fully complete the Work. This shall include all specialized workers that are required by the nature of the Work. Unless otherwise specifically provided herein, the costs of all labor are included in the Contract price. With respect to Work ordered for Con Edison, unless otherwise agreed to by Con Edison, Contractor shall employ on Work at the

construction site only union labor from building trades locals (affiliated with the Building & Construction Trades Council of Greater New York) having jurisdiction over the Work to the extent such labor is available. Where Contractor employs workers on sites where a permit to use or open a street (including excavating the street) is required and New York City Administrative Code Section 19-142, or its successor, (or a similar law, regulation, or code pertaining to sites located outside of New York City ("Similar Local Law")) is applicable, Contractor agrees that, pursuant to and in furtherance of the requirements of New York City Administrative Code Section 19-142, or its successor, (or the requirements of the Similar Local Law) and the terms and conditions of the permit: none but competent workers, skilled in the work required of them, shall be employed thereon; the prevailing scale of union wages shall be the prevailing wage for similar titles as established by the Comptroller of the City of New York pursuant to Section 220 of the New York State Labor Law (or as established by such other fiscal officer, as specified in Section 220 of the New York State Labor Law, for workers on permitted sites located outside of New York City to which a Similar Local Law applies), paid to those so employed, and Contractor shall pay that prevailing wage to workers so employed. These obligations of Contractor shall apply even though Con Edison may have obtained the permit. Contractor agrees to defend, save harmless and indemnify Con Edison, and its affiliates (including, but not limited to, O&R) and their respective trustees, directors, officers, employees, agents, representatives, successors and assigns from and against any and all liability arising in any way out of Contractor's failure to comply with the foregoing provisions of this paragraph. Whenever Contractor knows or believes that any actual or potential labor dispute is delaying or threatening to delay the timely performance of the Work, Contractor shall immediately give Con Edison notice of the dispute, including all relevant information concerning the dispute. If such notice is given orally, Contractor shall confirm it in writing within 24 hours. In the event that any labor dispute adversely affects the timely and efficient performance of the Work, Contractor shall exercise all rights and utilize all remedies available under applicable collective bargaining agreements and applicable federal and state laws to resolve the dispute and end the adverse effect on the Work, including but not limited to, seeking an injunction and filing an unfair labor practice charge.

B. The Immigration Reform and Control Act of 1986 Pub. L. No. 99-603 (the "Act") makes it illegal for an employer to hire or employ an illegal alien. The Act also makes it a legal requirement for employers to establish an employment verification system which includes the employer's checking specified documents to establish both an individual's identity and legal authorization to work. Contractor represents and covenants that Contractor has complied and will comply with all the requirements of the Act with respect to all persons assigned or employed by Contractor in the performance of the Work. Contractor agrees to defend, save harmless and indemnify Con Edison and its affiliates (including, but not limited to, O&R)

EXHIBIT NYSLOF-11

DATA DISPATCH

Gas utility M&A shakes up pipeline upgrade work, revenue streams

Thursday, August 22, 2019 12:45 PM ET

By Tom DiChristopher and Stephanie Tsao
Market Intelligence

Dealmaking in 2018 shook up an annual ranking of the natural gas utilities most actively repairing pipeline leaks, underscoring how companies view acquiring and replacing aging infrastructure as a viable path to rate expansion and revenue growth.

The 10 companies with the highest ratio of gas leak repairs to pipeline miles continue to be located mostly in the Northeast and mid-Atlantic, according to an analysis by S&P Global Market Intelligence. Safety concerns over obsolete pipeline materials and a number of deadly explosions in recent years have spurred utilities to fast track pipe replacement programs and accelerate deployment of leak detection technology.

One of the biggest changes to the annual ranking was the appearance of Elizabethtown Gas Co. in the top 10. The debut follows Southern Co.'s $1.7 billion sale of the Union, N.J.-headquartered utility and Maryland's Elkton Gas Co. to South Jersey Industries Inc., or SJI, in 2017.

The nearly 165-year-old company's arrival at number 4 also explains the absence of perennial top 10 finisher Pivotal Utility Holdings Inc. The Southern Co. subsidiary was essentially dismantled after SJI bought Pivotal's Elizabethtown and Elkton assets and NextEra Energy Inc. acquired its Florida City Gas business.

Gas utilities with the highest ratio of distribution main and service line leaks repaired in 2018

					Distribution main and service materials (%)						
Company	States of operation	Ratio of repaired leaks to miles	Number of repaired leaks	Main and line length (miles)	Unprotected steel (bare)	Unprotected steel (coated)	Cathodically protected steel (bare)	Cathodically protected steel (coated)	Plastic	Iron*	Other
Consolidated Edison Co. of New York Inc.	NY	1.62	12,351	7,646	17.59	1.39	0.00	7.03	59.10	13.12	1.75
Philadelphia Gas Works Co.	PA	0.97	5,735	5,933	8.22	10.09	0.00	11.06	45.82	24.73	0.02
Memphis Light Gas and Water Division	TN	0.75	6,028	8,054	0.00	0.00	0.00	45.31	54.56	0.12	0.00
Elizabethtown Gas Co.	NJ	0.65	3,549	5,483	0.54	0.33	0.69	20.73	64.72	8.04	4.92
Baltimore Gas and Electric Co.	MD	0.60	8,269	13,731	5.48	0.00	0.00	25.57	59.28	8.14	1.51
Boston Gas Co.	MA	0.58	6,379	10,921	12.15	5.56	1.12	10.16	47.62	16.01	7.35
Brooklyn Union Gas Co.	NY	0.53	4810	9,014	5.56	0.00	0.00	15.85	54.51	14.12	9.93
Mountaineer Gas Co.	WV	0.47	3,152	6,739	25.20	2.24	0.37	16.69	55.50	0.00	0.00
Southern California Gas Co.	CA	0.43	43,419	101,042	3.21	13.88	0.00	26.13	56.79	0.00	0.00
Virginia Natural Gas Inc.	VA	0.43	4,763	11,111	0.29	3.83	1.80	18.95	74.97	0.01	0.15

Data compiled Aug. 6, 2019.
* Includes cast and ductile iron.
Includes companies with at least 5,000 miles of distribution mains and service lines.
Service line length estimated from count of lines and average length.
Source: S&P Global Market Intelligence

New top 5 entrant extends robust replacement program through M&A

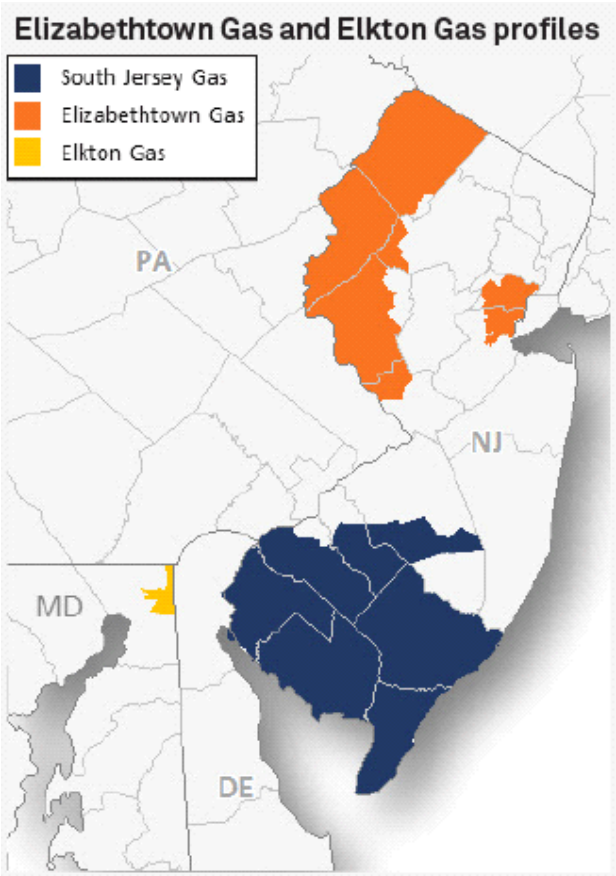
Elizabethtown Gas repaired 3,549 leaks in 2018, representing a repaired-leak ratio of 0.65 along nearly 5,500 miles of distribution mains and service lines. That was higher than any annual ratio attributable to Pivotal Utility Holdings going

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back to 2013, when S&P Global Market Intelligence first analyzed data from the U.S. Pipeline and Hazardous Materials Safety Administration for utilities with at least 5,000 miles of pipeline.

SJI had already been investing heavily in pipe replacement in its existing service territory in southern New Jersey when it struck the Elizabethtown Gas deal. Because of those investments, SJI expects to complete bare steel and iron pipe replacement in its legacy service territory, where it operates as South Jersey Gas Co., by 2021.

The acquisition gave SJI the opportunity to tee up continued revenue growth by replacing the iron pipes and unprotected steel lines that make up 8% and nearly 1%, respectively, of the Elizabethtown Gas system. That opportunity was particularly attractive to SJI because the assets were within a state where the utility already had experience executing an accelerated replacement program.



Source: South Jersey Industries

"When [SJI] evaluated the acquisition they saw that Elizabethtown Gas still had a significant amount of replacement work needed," Christie McMullen, president of Elizabethtown Gas, said in an interview. "That really ... created an opportunity to take the best practices that are there at South Jersey Gas for infrastructure replacement processes, to apply that in a regulatory environment that they were familiar with operating in."

The New Jersey Board of Public Utilities in June authorized Elizabethtown Gas to replace 250 miles of pipeline and make other system upgrades at a cost of $300 million. SJI is negotiating with the board for a $65 million base rate increase to cover the cost of the improvements.

Despite its divestment from Elizabethtown Gas, Southern Co. is nevertheless represented on this year's list through its subsidiary Virginia Natural Gas Inc. The Virginia Beach-headquartered utility broke into the top 10 in 2018 with a leak-repair ratio of 0.43.

Virginia Natural Gas and Southern Co.'s Illinois subsidiary Nicor Gas both have rate riders that allow them to quickly recover capital expenditures from programs to replace cast iron, bare steel and older plastic infrastructure or expand gas distribution systems, the company noted in its 2018 annual report. Southern Co. expects to spend $408 million on these programs in 2019.

Climate activism impacting pipe upgrade attitudes

As utilities continue to aggressively tackle gas leaks, climate activists are focusing more squarely on sidelining natural gas, the cleanest burning fossil fuel but still a major contributor to global warming, and potentially crimping demand and eliminating some opportunities to repair infrastructure.

Bloomberg Philanthropies announced the Beyond Carbon campaign in March to build on its work reducing U.S. coal consumption and begin shrinking the role of oil and gas in the nation's energy mix.

Berkeley, Calif., adopted a ban on new natural gas hookups in July, and similar measures are being considered in other cities. Most of the statutes under consideration would impact the Golden State, but analysts said cities with ambitious climate goals and aging pipeline systems are also among the most likely to consider similar measures.

Local lawmakers in Massachusetts' Brookline, Cambridge and Lexington communities already plan to introduce Berkeley-style gas bans. Brookline and Lexington are serviced by National Grid PLC subsidiary Boston Gas Co., which had the sixth highest leak-repair ratio in 2018. About one-third of Boston Gas' system is still comprised of unprotected steel and iron pipe, leaving ample opportunity for revenue-boosting replacement.

There is also a push for building electrification underway in Philadelphia, where questions about the financial viability of

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Philadelphia Gas Works Co. have been swirling for years. The nation's largest municipal gas utility had the second-highest leak-repair ratio for the sixth year in a row, as it faces persistent pressure from state regulators to address a lengthy backlog of leaks.

Philadelphia Gas Works still has the lowest percentage of plastic pipeline and the highest percentage of iron and unprotected steel in its system among the top 10. Corrosion on its pipelines accounted for nearly 40% of leak repairs in the system.

Leak causes for gas utilities with the highest ratio of distribution main and service line leaks repaired in 2018

Company	Number of repaired leaks	Cause of leaks (%)							
		Corrosion	Equipment	Operations	Excavation	Outside force	Natural forces	Material or welds	Other*
Consolidated Edison Co. of New York Inc.	12,351	48.56	20.57	0.15	1.98	0.29	1.85	0.65	25.94
Philadelphia Gas Works Co.	5,735	39.56	14.25	0.17	2.48	0.71	41.74	0.14	0.94
Memphis Light Gas and Water Division	6,028	0.80	91.79	0.33	4.79	0.40	0.05	1.38	0.46
Elizabethtown Gas Co.	3,549	8.31	64.58	2.37	5.55	1.52	1.58	16.03	0.06
Baltimore Gas and Electric Co.	8,269	16.33	6.64	1.09	3.34	1.03	4.84	62.84	3.91
Boston Gas Co.	6,379	33.12	6.15	0.11	7.15	0.47	7.18	0.55	45.27
Brooklyn Union Gas Co.	4,810	11.08	17.88	0.00	5.22	0.79	4.20	0.10	60.73
Mountaineer Gas Co.	3,152	75.73	0.51	0.03	5.04	0.86	3.93	0.00	13.90
Southern California Gas Co.	43,419	24.71	37.56	10.87	7.82	1.54	2.40	13.98	1.12
Virginia Natural Gas Inc.	4,763	22.04	39.39	6.30	5.19	1.18	0.63	25.28	0.00

Data compiled Aug. 6, 2019.
* Other includes material and construction defects.
Includes companies with at least 5,000 miles of distribution mains and service lines.
Source: S&P Global Market Intelligence

Across the top 10, corrosion and problems with equipment such as pipe fittings and valves continued to be the top causes of leaks, on average each accounting for about 30% of the leaks companies tackled. The percentage of equipment-related issues among the top 10 was up significantly from last year and just edged out corrosion as the most prevalent cause of leaks repaired in 2018.

Safety concerns keep leak detection on the front burner

Few industry analysts expect gas bans to spread like wildfire, and the Berkeley statute only impacts new buildings, leaving opportunities to replace pipelines that serve existing customers. Meanwhile, fatal explosions on distribution systems in major U.S. cities have kept pipeline safety top of mind among ratepayers and regulators.

Consolidated Edison Co. of New York Inc. once again topped the list, a position it has held since 2014 when a deadly blast in New York City's Harlem neighborhood prompted the company to shore up its gas leak detection efforts.

In 2018, the Consolidated Edison Inc. subsidiary reduced the percentage of unprotected bare steel to just under 19%, knocking a full percentage point off the 2017 total and keeping with the pace of the last several years. Iron pipes accounted for just over 13% of its system, down about half a percentage point year over year.

Overall, the ConEd utility's leak-repair ratio climbed to 1.62, one of the highest on record for its 12,351-mile system. ConEd's 2018 capital requirements for its gas operations surpassed $1 billion in 2018 and are projected top $1.1 billion in 2020. It plans to replace 100 miles of pipe per year in 2021, up from more than 90 miles in 2018 and 80 miles in 2017.

The list of companies with the lowest leak-repair ratios was once again dominated by companies with systems comprising entirely or mostly modern materials that are less likely to leak. Many of those are located in the Midwest and western U.S.

Portland, Ore.-based Northwest Natural Gas Co., which only transports gas through plastic and cathodically protected steel pipes, repaired just 804 leaks along its 23,824-mile system, posting the lowest ratio at 0.03. Companies among the bottom 10 with some of the highest percentages of leak-prone pipes, including West Texas Gas Inc. and New York State Electric & Gas Corp., continued to strip unprotected steel and iron mains and lines from their systems in 2018, driving

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the percentage of their pipeline mileage comprised of those materials into the low single digits.

Gas utilities with the lowest ratio of distribution main and service line leaks repaired in 2018

Company	States of operation	Ratio of repaired leaks to miles	Number of repaired leaks	Main and line length (miles)	Distribution main and service materials (%)						
					Unprotected steel (bare)	Unprotected steel (coated)	Cathodically protected steel (bare)	Cathodically protected steel (coated)	Plastic	Iron*	Other
Puget Sound Energy Inc.	WA	0.07	1,731	25,937	0.00	0.00	0.00	23.29	76.71	0.00	0.00
Colorado Springs Utilities	CO	0.06	352	5,428	4.69	0.04	0.00	19.09	74.79	0.00	1.39
Rochester Gas and Electric Corp.	NY	0.06	564	9,018	0.48	0.50	0.08	39.24	58.68	0.01	1.02
New Mexico Gas Co. Inc.	NM	0.06	867	15,491	0.00	0.00	0.66	42.00	57.34	0.00	0.00
Wisconsin Gas LLC	WI	0.05	1,079	20,182	0.00	0.00	0.00	27.91	71.20	0.00	0.89
ENSTAR Natural Gas Co.	AK	0.05	295	5,998	0.00	0.00	0.02	12.99	86.58	0.00	0.41
New York State Electric & Gas Corp.	NY	0.04	369	8,383	1.52	2.72	0.08	31.30	63.43	0.10	0.85
West Texas Gas Inc.	LA, NM, OK, TX	0.04	246	5,853	2.12	1.01	2.18	13.71	80.98	0.00	0.00
Wisconsin Electric Power Co.	MI, WI	0.04	632	17,030	0.00	0.00	0.00	23.40	76.60	0.00	0.00
Northwest Natural Gas Co.	OR, WA	0.03	804	23,824	0.00	0.00	0.00	38.47	61.53	0.00	0.00

Data compiled Aug. 6, 2019.
* Includes cast and ductile iron.
Includes companies with at least 5,000 miles of distribution mains and service lines.
Service line length estimated from count of lines and average length.
Source: S&P Global Market Intelligence

But recent regulatory reviews in New York show that there are still risks, even when using modern materials such as plastic. ConEd, along with National Grid subsidiary Brooklyn Union Gas Co. — which had the seventh highest leak-repair ratio in 2018 — face potential fines over allegations they failed to properly oversee the work of contractors who hired unqualified plastic pipe fusers to complete New York gas pipeline projects.

Rapid deployment of leak detection technology can also create challenges for gas utilities. Dallas-based Atmos Energy Corp. reported second-quarter operating costs rose faster than anticipated as the company hired additional workers to process leak detection data and dedicated more time to processing unfamiliar information streams. Like ConEd, Atmos ramped up spending on pipe replacement following a deadly 2018 blast in Dallas.

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EXHIBIT NYSLOF-12

Date of Request: July 11, 2019 Request No. DPS-865
Due Date: July 22, 2019 NG Request No. NG-1149

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
Case Nos. 19-G-0309 & 19-G-0310
Gas Utilities Rates

Request for Information

FROM: DPS Staff, Michael Pasinella

TO: National Grid, Gas Safety Panel

SUBJECT: Northeast Supply Enhancement (NESE) Project

Request:

Note: In all interrogatories, all requests for workpapers or supporting calculations shall be construed as requesting any Word, Excel or other computer spreadsheet models in original electronic format with all formulae intact and unlocked.

For both KEDNY and KEDLI:

1. Should the NESE project be postponed, or denied:

   a. How would the companies adjust their labor forces?

   b. Were any of the requested incremental full-time equivalent positions dedicated to work related to the NESE project? If so, describe the activity and detail how many FTEs would be allocated to each activity if NESE is permitted.

   c. Could labor resources be re-deployed to offset any of the proposed work in the rate and subsequent data years? If so, specifically what projects would be offset and what are the respective associated costs?

2. If non-company personnel were forecasted to work on the NESE project, what would the cost reductions be in the rate and subsequent data years?

3. How would the postponement or denial of the NESE project impact the company's operations? For example, will the company need to install more CNG facilities? If so, how will that impact the company's operations and maintenance programs? Will there be other changes to the operations and maintenance programs?

Response:

1. a. Assuming an extended delay or denial of the NESE project, KEDNY's contractor labor force will be reduced by approximately one to three crews, and KEDLI's contractor labor force will be reduced by approximately 13 crews due to the reduction in capital programs described in the Supplemental Direct Testimony of the Companies' respective Gas Infrastructure and Operations Panels that was filed on June 10, 2019 ("GIOP Supplemental Testimony"). The Companies are continuing to evaluate future labor impacts should the NESE project not be approved.

   b. There are no incremental FTEs dedicated to the NESE project. As is stated in the GIOP Supplemental Testimony, the reduction in resources required to address Company programs would result in a reduction of incremental contractor inspector FTEs.

   c. The GIOP Supplemental Testimony provides the projects and programs that would be offset and associated cost adjustments.

2. There are no contractor labor resources dedicated to work on the NESE project. Anticipated reductions to the Companies' contractor labor needs are provided in response to part 1, above.

3. The impacts of the postponement or denial of the NESE project on the Companies' capital and O&M programs are set forth in the GIOP Supplemental Testimony. The Companies are not currently planning to deploy additional CNG and are continuing to analyze options to address supply constraints resulting from a delay or denial of the NESE project.

Name of Respondent:
Frank Prost
Melissa Mancini
Elizabeth Arangio

Date of Reply:
July 22, 2019